LABORATORY OF AUTOMATION SYSTEMS

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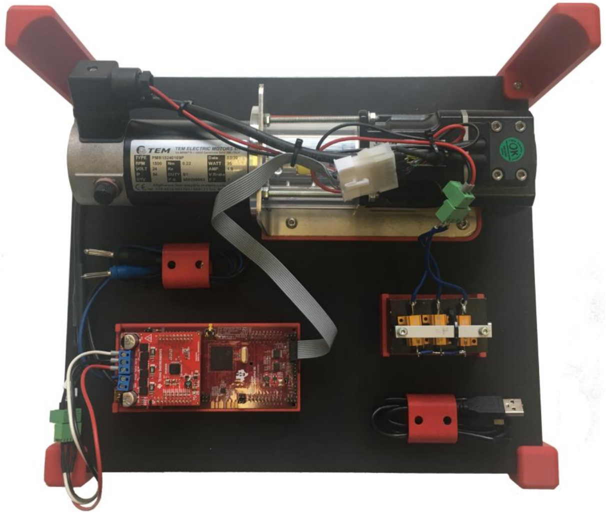
Today Goals:

- Connect setup with PC
- First Project
- Overview of Microprocessor Peripherals
  - Digital-IO
  - ADC
  - PWM

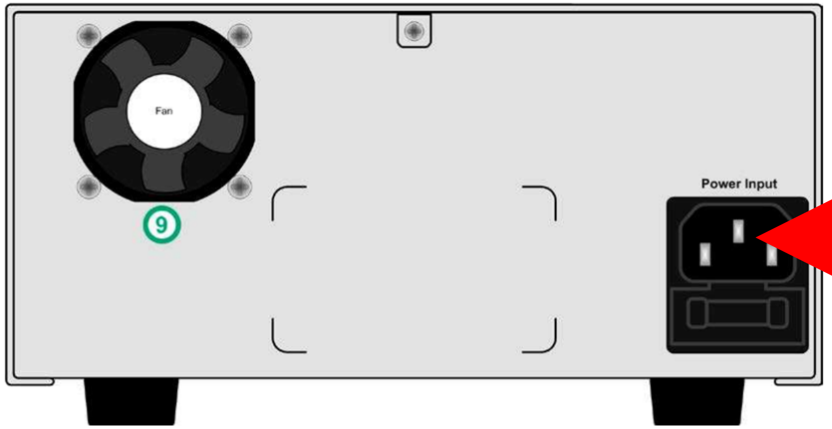
HOW TO CONNECT

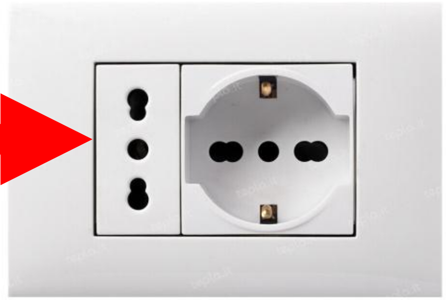
1: Power Supply

2: Power Supply cable



3: Setup





Connect cable

Check/Set Voltage and Current Values

by pressing the **preset** button we can access the current/voltage menu **V/C**
We can change voltage and current.
Here we can choose our output voltage and current limit

By turning the knobs we can change the voltage and current

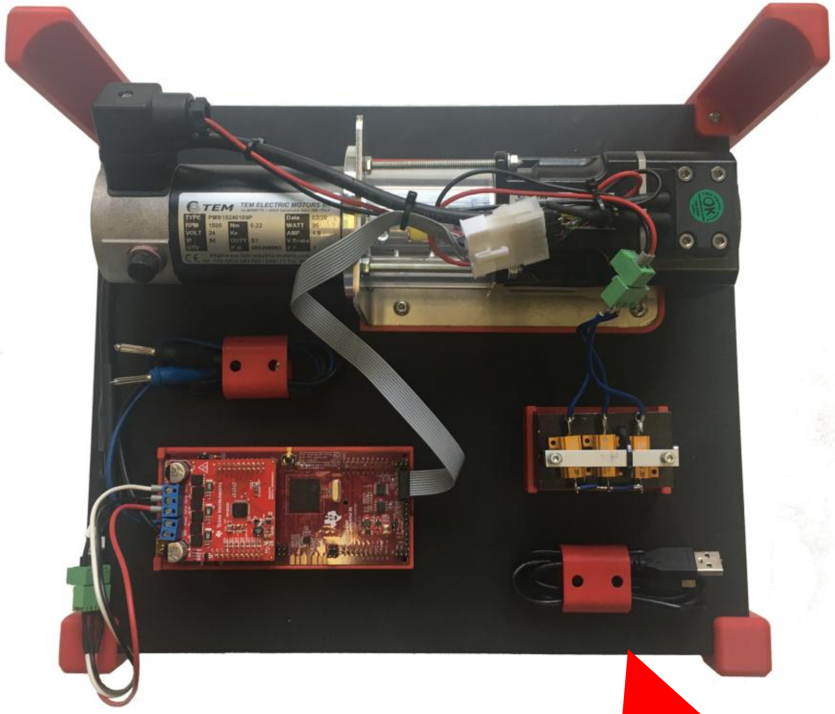
For the moment set: 24V – 2.5A

Rotary Button

Check/Set Voltage and Current Values

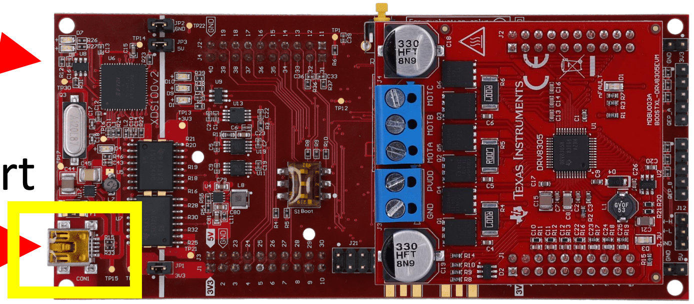
by pressing the preset button we can access the current/voltage menu **OVP/OCP** Over Voltage Protection/Over Current Protection
We can change voltage and current limit.
If we exceed this limit, the Power Supply disconnects the output and report an error.

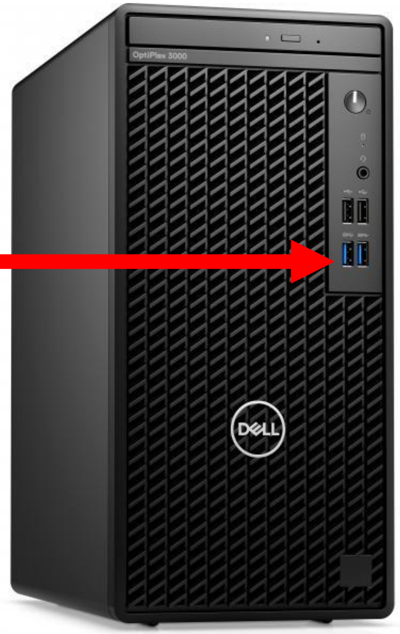
By turning the knobs we can change the voltage and current



LED On

Mini-USB Port



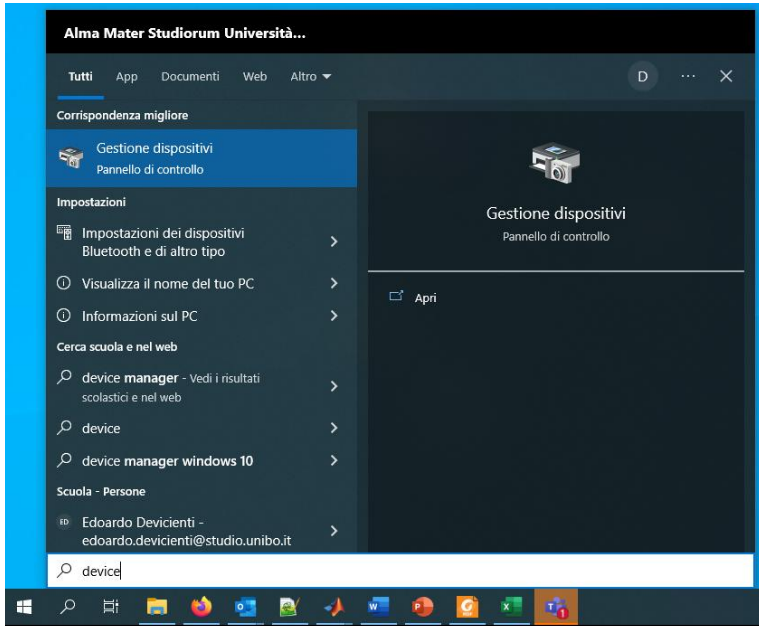


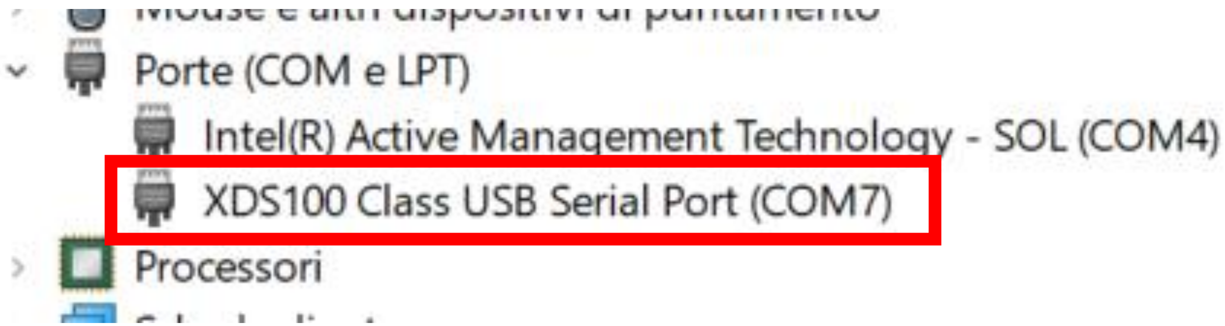
1: Take the USB Cable

2: Connect Board-PC

We need to find the **com port** used by the board:

- Open *Device Manager*
- Select *Porte (COM e LTP)*
- Remember your COM port, in the figure **COM7**

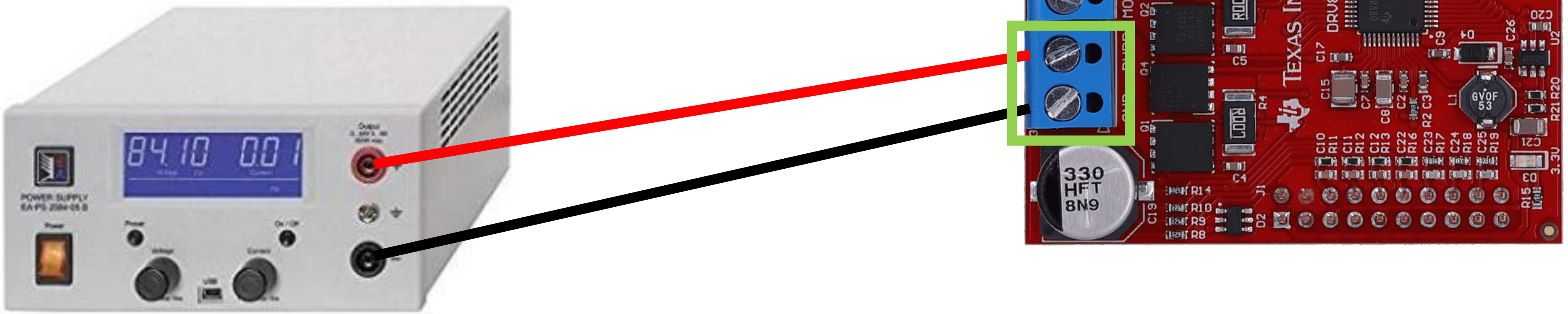




Please wait for my check

!! ATTENTION !!

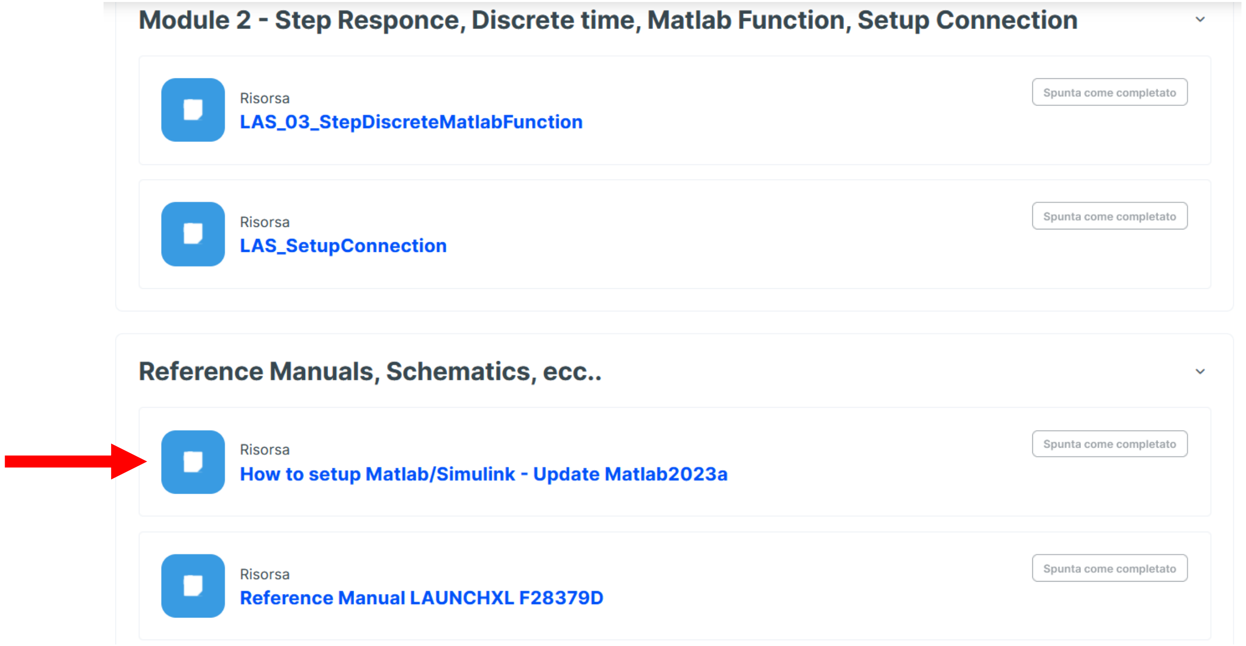
Red/Black +/-

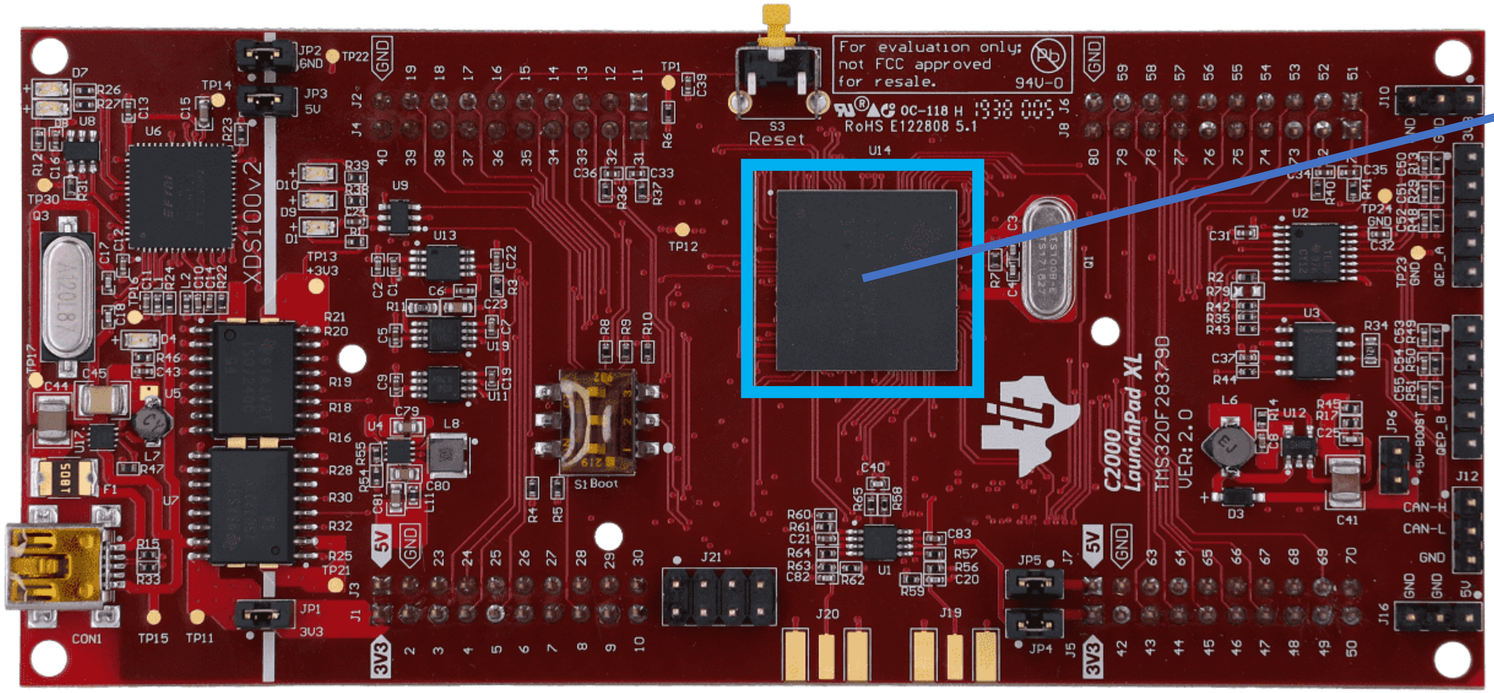


Let’s create now a new Simulink Project for our microprocessor.

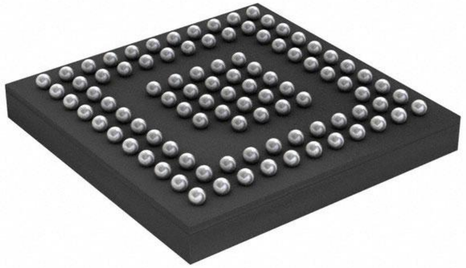
You can get the guide from virtuale or Teams: **MatlabSetupForTexas.pdf**

Start from page 11





BOTTOM VIEW



We have > 300pin:

- Power supply
- Crystal (clock)
- GPIO (general purpose I/O)
- ADC (Analog to digital converter)
- PWM (Pulse Width Modulation)
- USART
- Ecc….

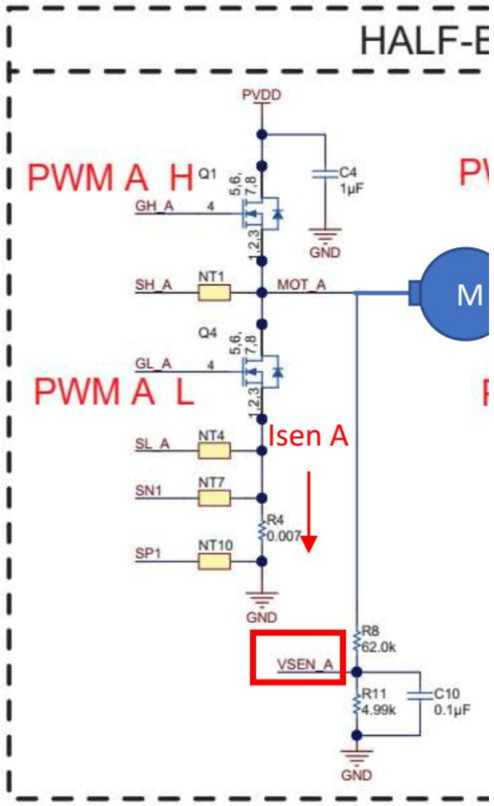
A pin can be programmed to have different functions

As an example, if we look the driving pin of PWM_A_H

H-Bridge

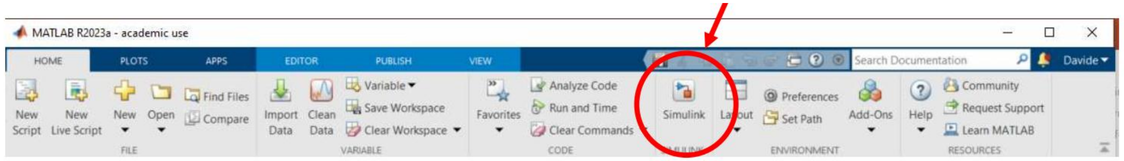
Table 8. F28379D LaunchPad Pin

Mux Value				J8 Pin
X	2	1	0	
	PWM A H	EPWM4A	GPIO6	80
	PWM A L	EPWM4B	GPIO7	79
	PWM B H	EPWM5A	GPIO8	78
	PWM B L	EPWM5B	GPIO9	77
		EPWM6A	GPIO10	76
		EPWM6B	GPIO11	75
OUTPUTXBAR3 (1)			GPIO14	74
OUTPUTXBAR4 (1)			GPIO15	73
			DAC3	72
			DAC4	71



We can drive it as a PWM o GPIO, more if we look datasheet…

GPIO6	0, 4, 8, 12	A6	166	–	I/O	General-purpose input/output 6
EPWM4A	1				O	Enhanced PWM4 output A (HRPWM-capable)
OUTPUTXBAR4	2				O	Output 4 of the output XBAR
EXTSYNCOUT	3				O	External ePWM synch pulse output
EQEP3A	5				I	Enhanced QEP3 input A
CANTXB	6				O	CAN-B transmit


MATLAB R2023a - academic use
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New Script
New Live Script
New
Open
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Compare
Import Data
Clean Data
Variable
Save Workspace
Clear Workspace
Favorites
Analyze Code
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Clear Commands
Simulink
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Community
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FILE
VARIABLE
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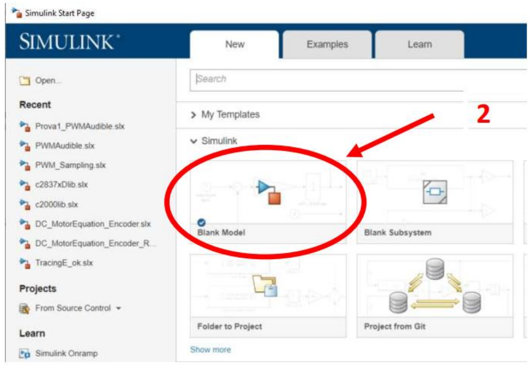
ENVIRONMENT
RESOURCES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PWMAudible.slx
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c2837xDlib.slx
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Blank Subsystem
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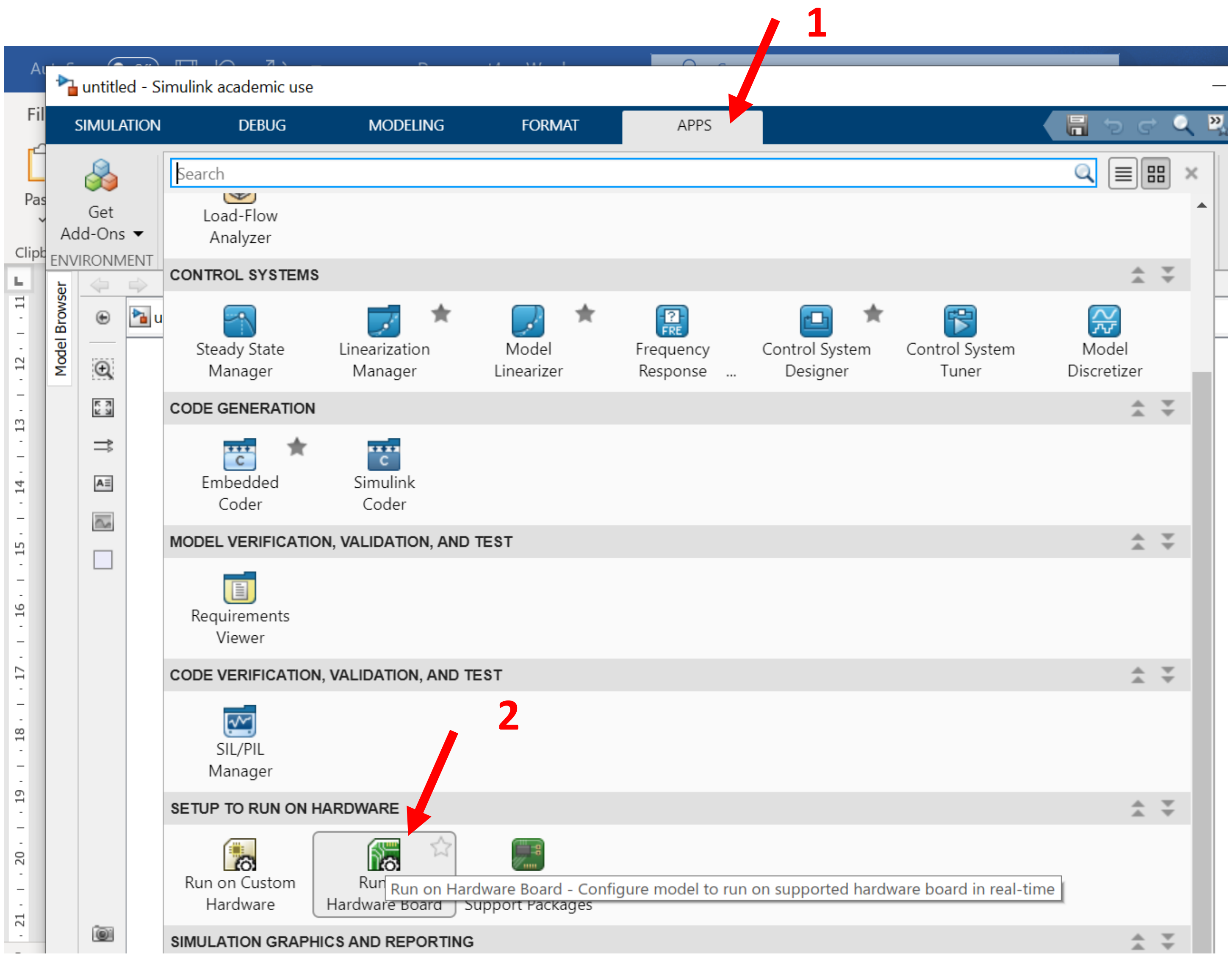
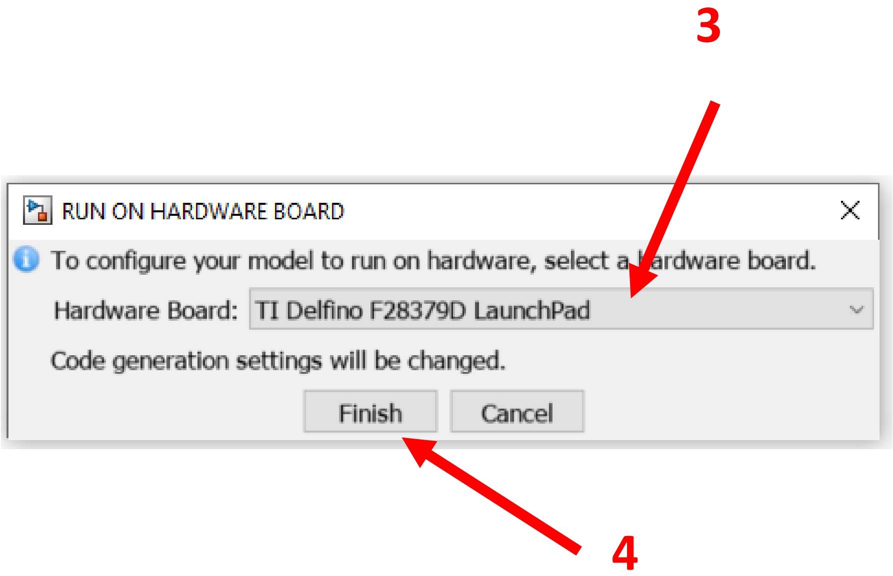
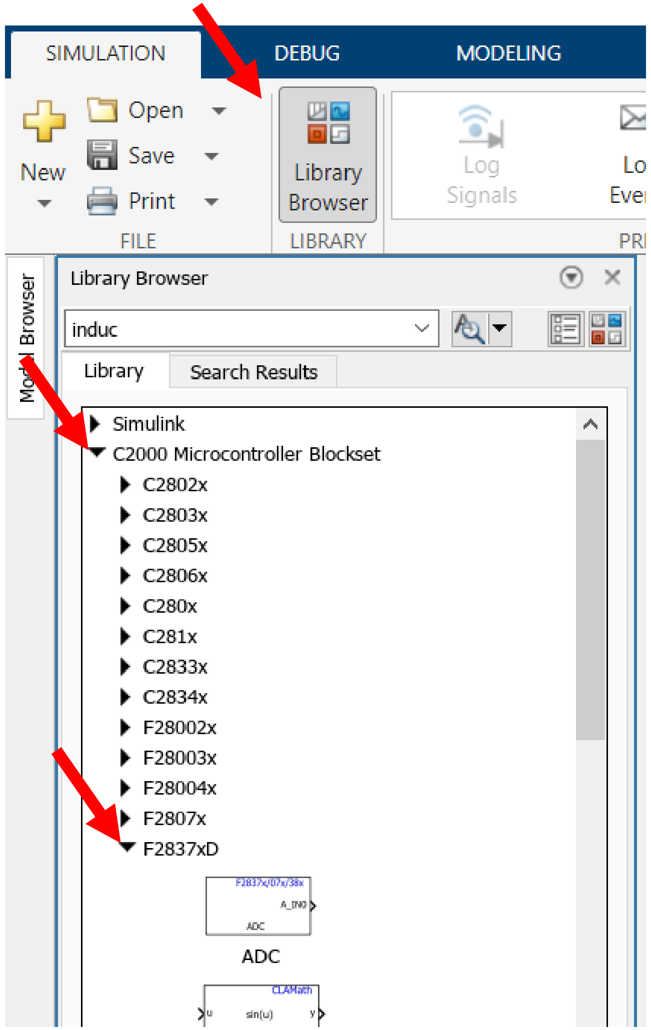


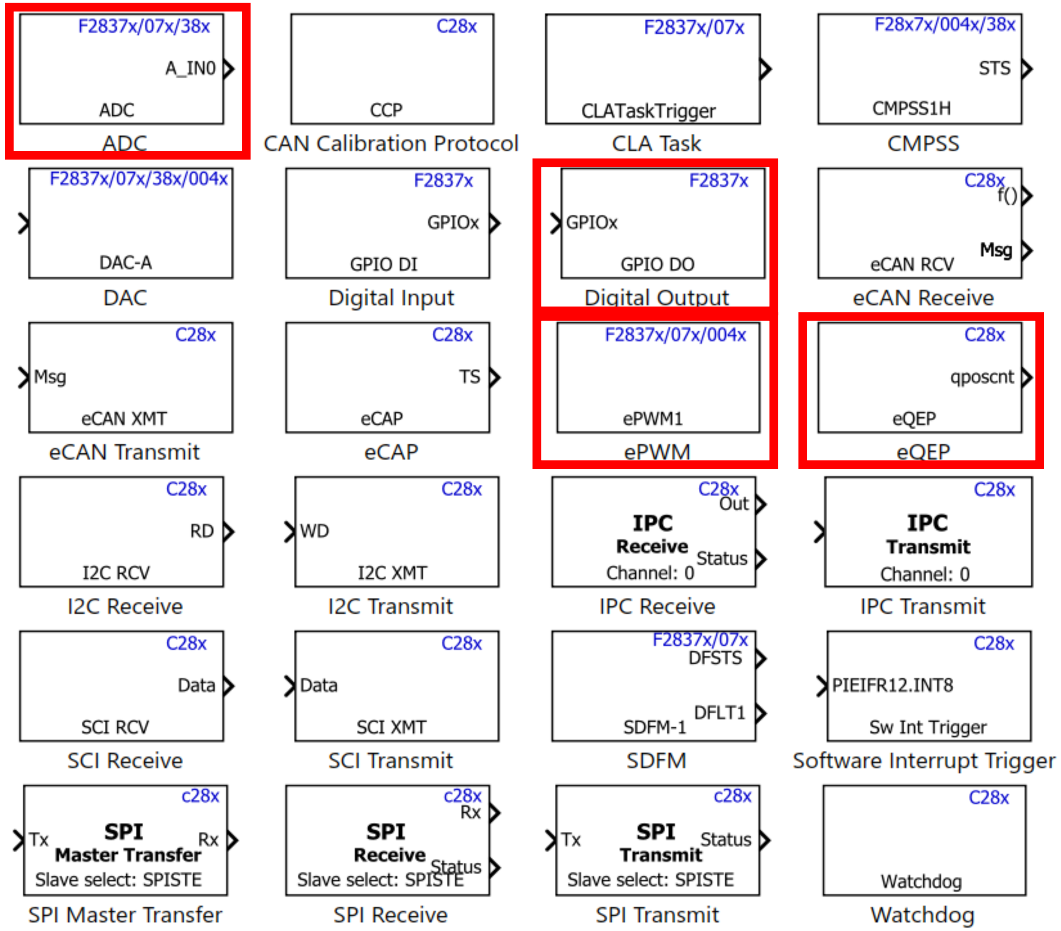
1
untitled - Simulink academic use
SIMULATION
DEBUG
MODELING
FORMAT
APPS
Get Add-Ons
ENVIRONMENT
Search
Load-Flow Analyzer
CONTROL SYSTEMS
Steady State Manager
Linearization Manager
Model Linearizer
Frequency Response ...
Control System Designer
Control System Tuner
Model Discretizer
CODE GENERATION
Embedded Coder
Simulink Coder
MODEL VERIFICATION, VALIDATION, AND TEST
Requirements Viewer
CODE VERIFICATION, VALIDATION, AND TEST
SIL/PIL Manager
2
SETUP TO RUN ON HARDWARE
Run on Custom Hardware
Run on Hardware Board
Run on Hardware Board - Configure model to run on supported hardware board in real-time
Support Packages
SIMULATION GRAPHICS AND REPORTING



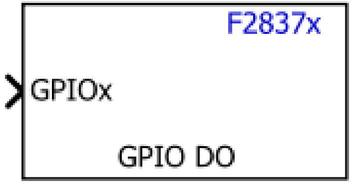
3
RUN ON HARDWARE BOARD
To configure your model to run on hardware, select a hardware board.
Hardware Board: TI Delfino F28379D LaunchPad
Code generation settings will be changed.
Finish
Cancel
4

Here the Simulink blocks that we are going to use:





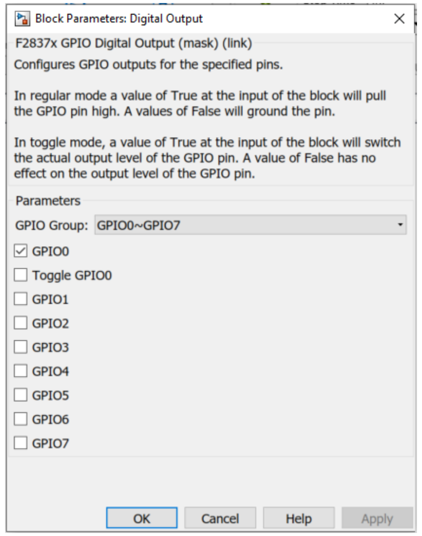
Digital output

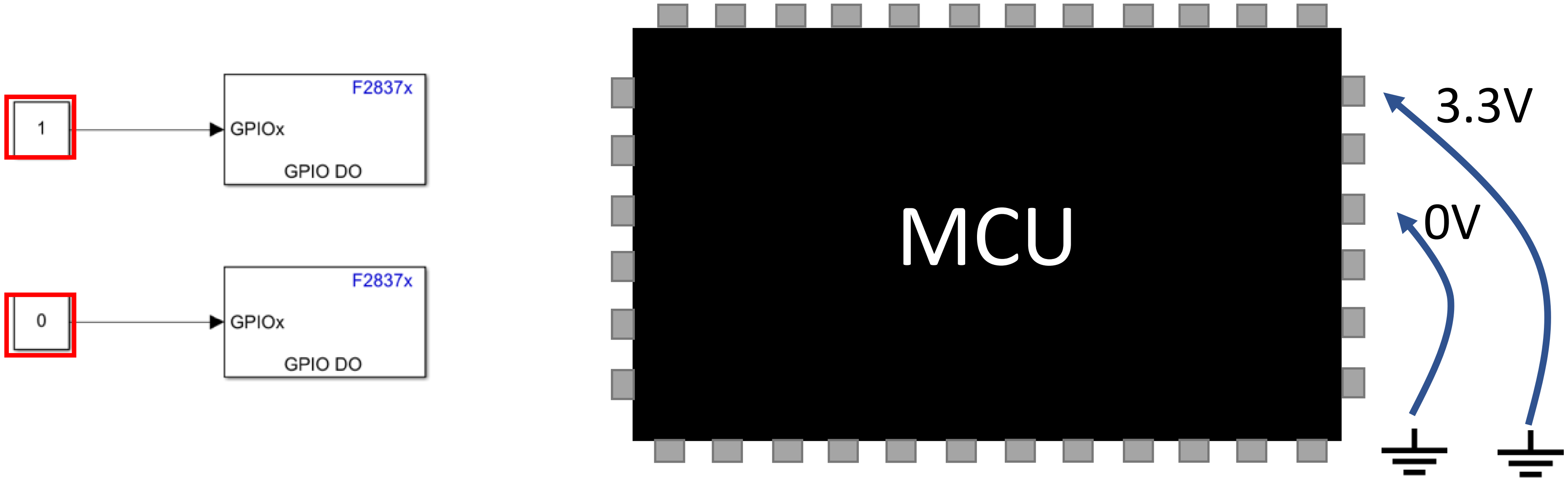


Digital Output

With this block, we can drive a pin of our microcontroller (MCU) as a digital output:

- with a 1 on input, we will have 3.3V on the pin
- with a 0 on input, we will have 0V on the pin

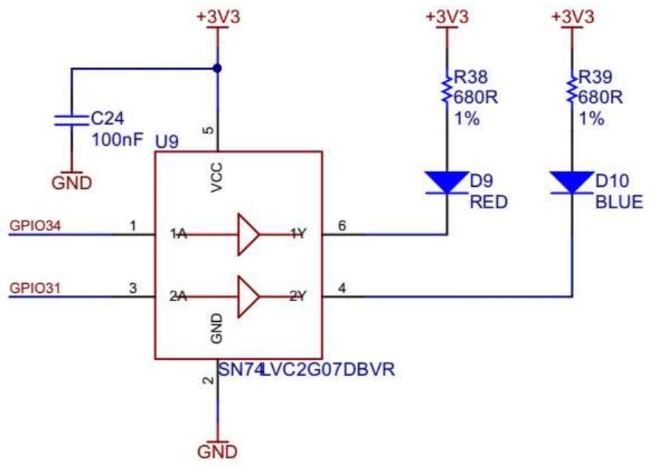




Digital output

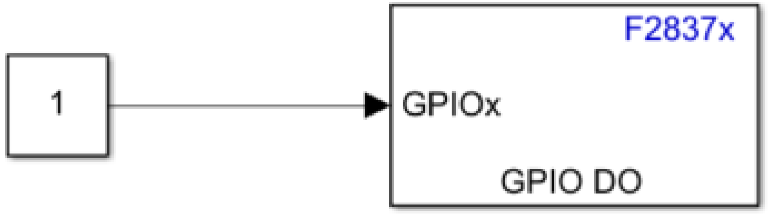
In our projects:
- LED
- H-Bridge Enable

From schematics:



GPIO34	Led RED
GPIO31	Led BLUE

0V o 3.3V?



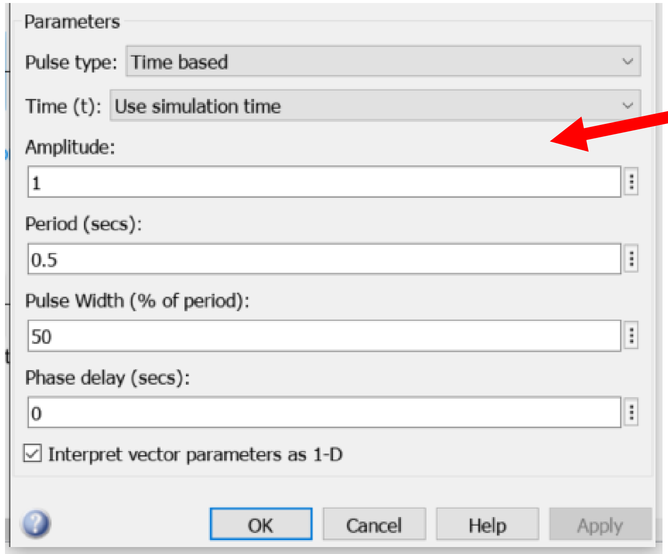
Micro Datasheet

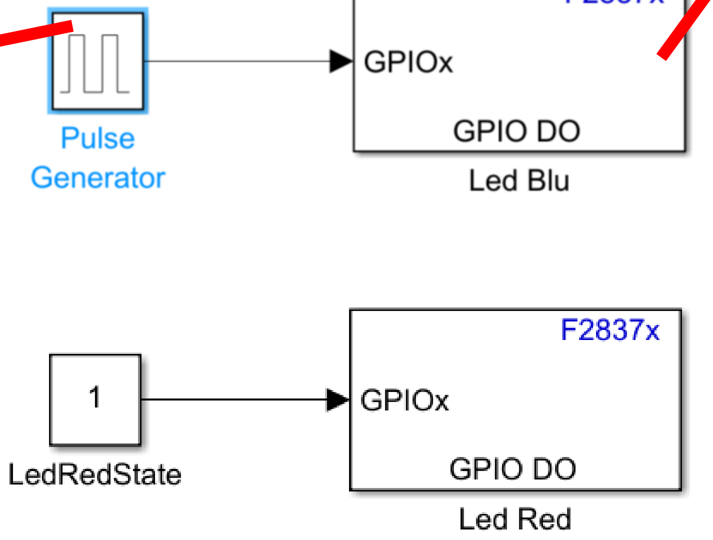
Eval Schematic

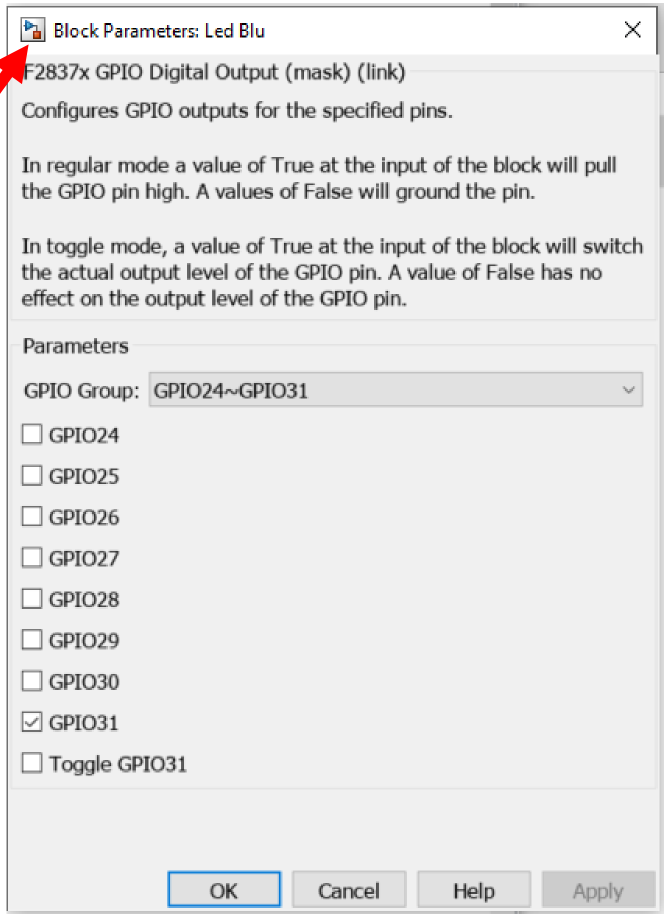
Digital output

GPIO31	Led BLUE
GPIO34	Led RED

Example: Blink blue LED, manual Red LED

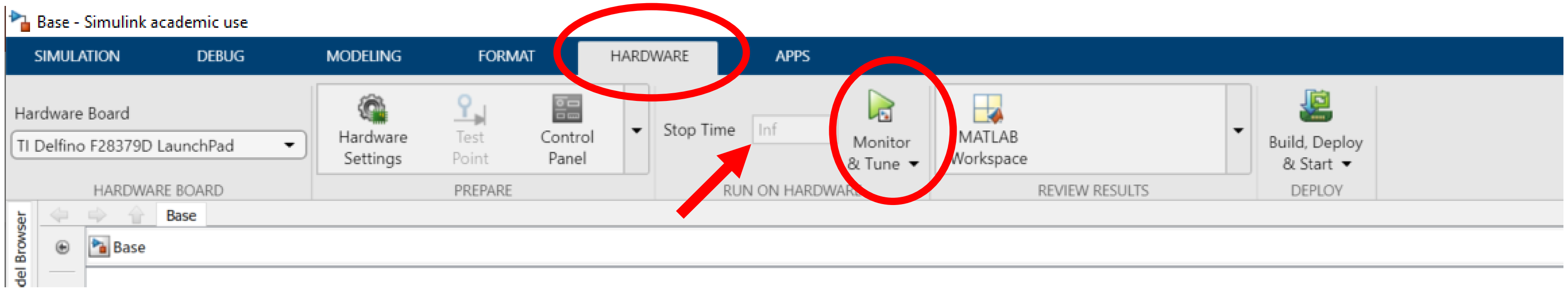




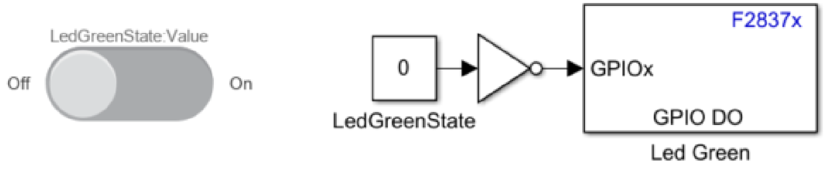


Digital output

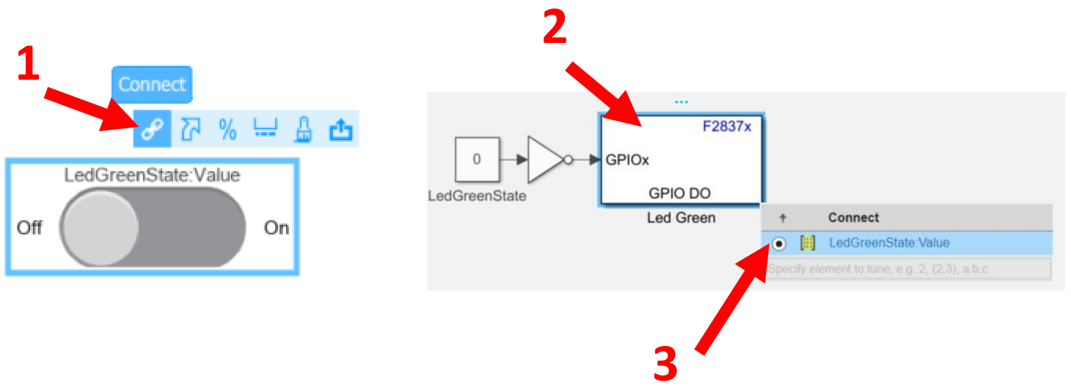
How to download scheme:



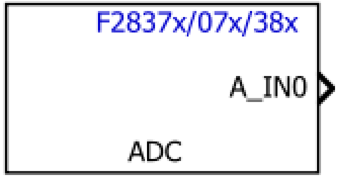
SLIDER SWITCH:



To Connect slider with Constant



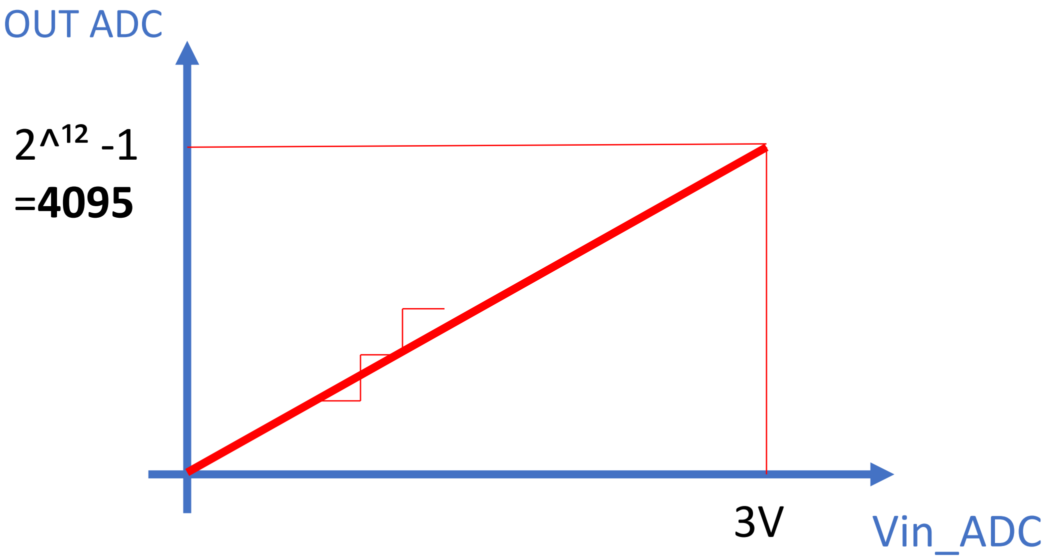
ADC



ADC

Analog to Digital Converter

In our board we have a **12bit ADC** with a reference voltage of **3V**.
An input voltage in a range from 0 to 3V is mapped into a number from 0 to 4095 ($2^{12}-1$)



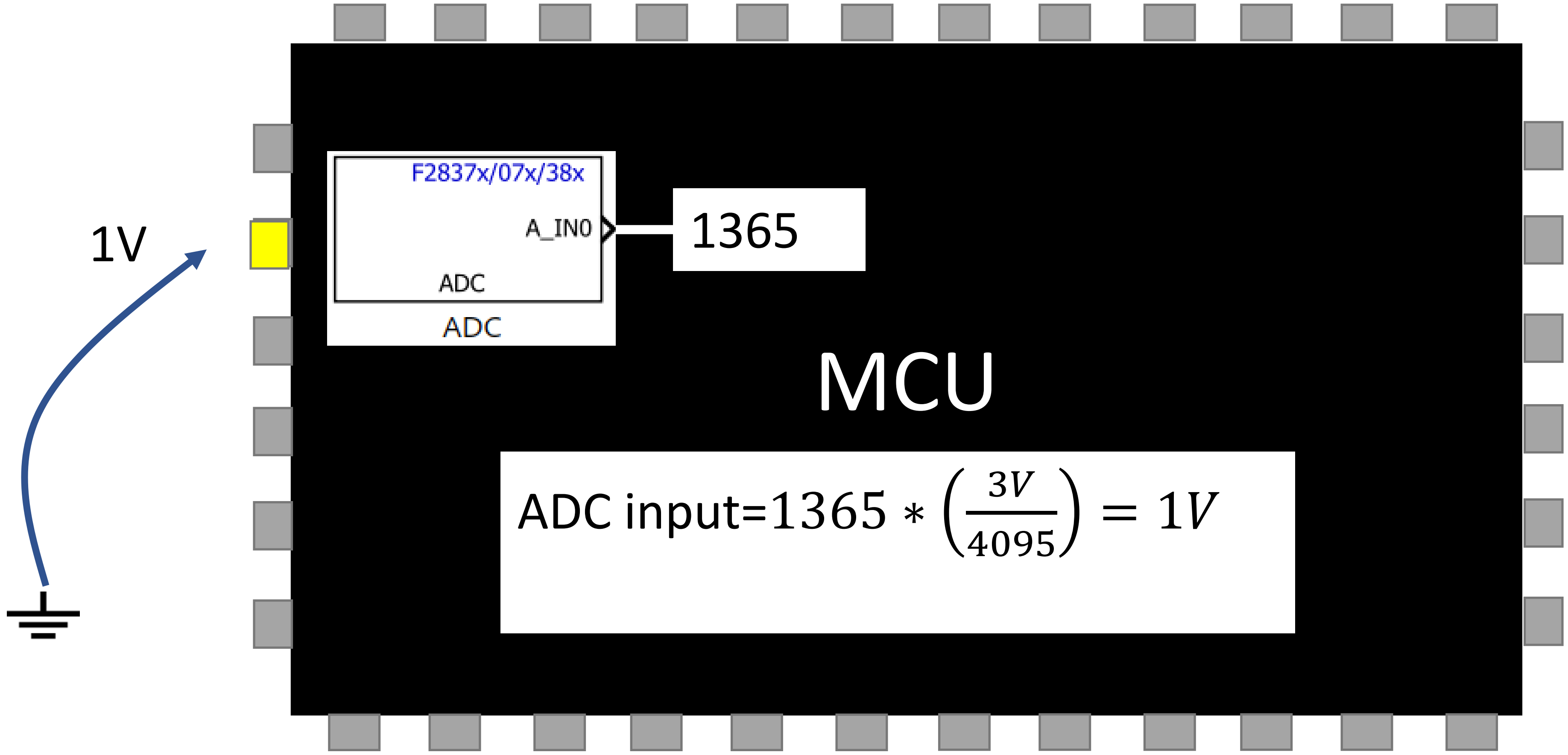
So, if we have in input 1V, our ADC output will be:

OUT ADC=$1V * \left(\frac{4095}{3V}\right) = 1365$ (int!)

We have 1365 in ADC output, ADC input is

ADC input=$1365 * \left(\frac{3V}{4095}\right) = 1V$

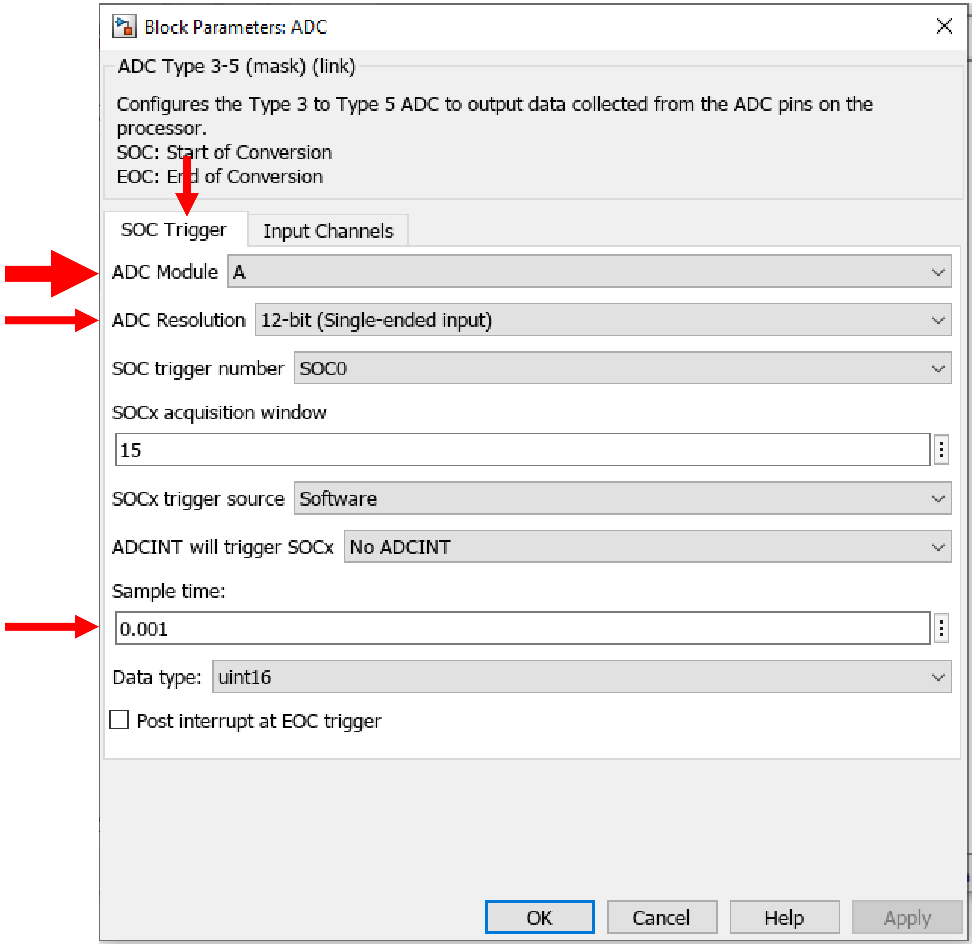
ADC

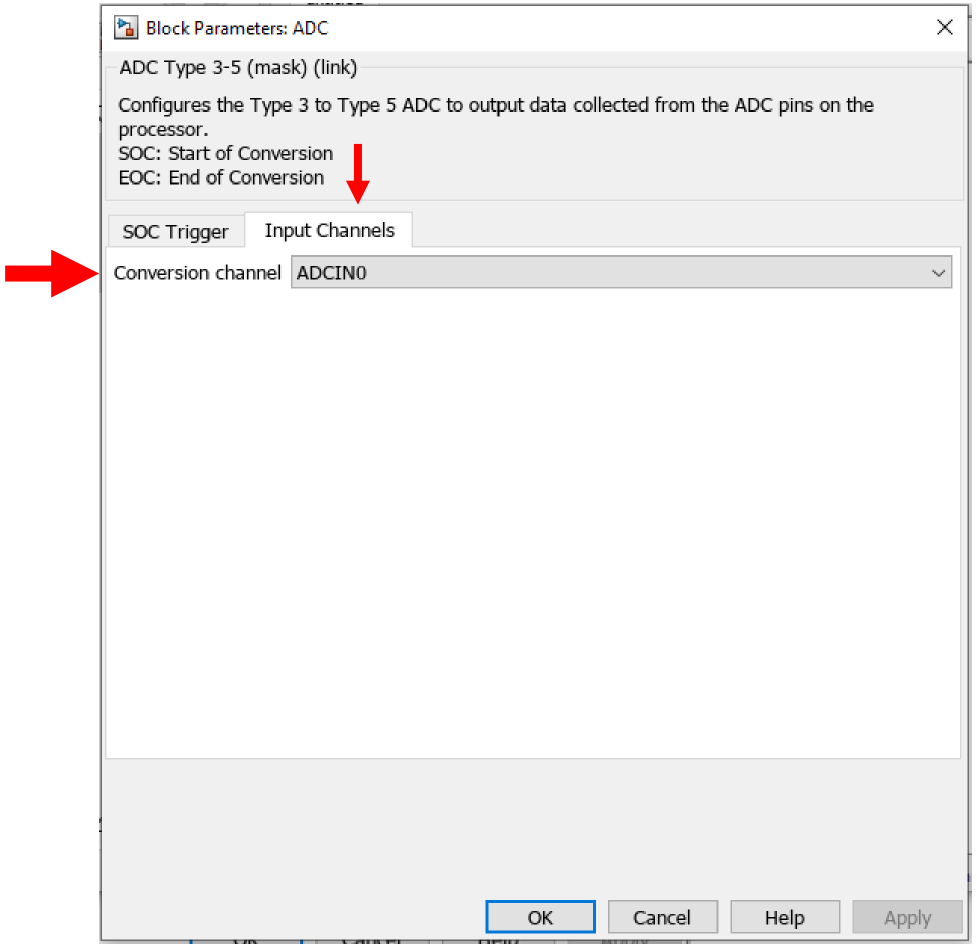


1V
F2837x/07x/38x
A_IN0
ADC
ADC
1365
MCU
ADC input=1365 $* \left(\frac{3V}{4095}\right) = 1V$

ADC


F2837x/07x/38x
A_IN0
ADC


Block Parameters: ADC
ADC Type 3-5 (mask) (link)
Configures the Type 3 to Type 5 ADC to output data collected from the ADC pins on the processor.
SOC: Start of Conversion
EOC: End of Conversion
SOC Trigger
Input Channels
ADC Module A
ADC Resolution 12-bit (Single-ended input)
SOC trigger number SOC0
SOCx acquisition window
15
SOCx trigger source Software
ADCINT will trigger SOCx No ADCINT
Sample time:
0.001
Data type: uint16
Post interrupt at EOC trigger
OK
Cancel
Help
Apply

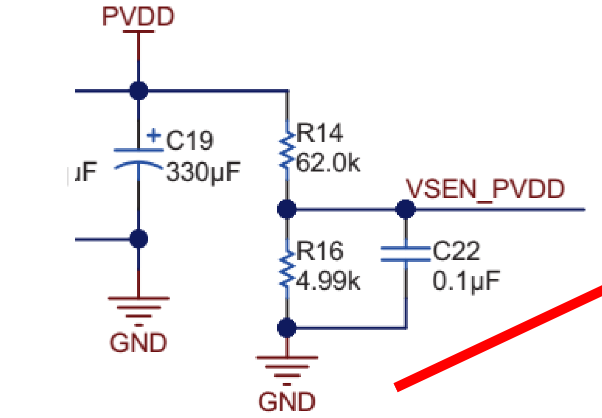

Block Parameters: ADC
ADC Type 3-5 (mask) (link)
Configures the Type 3 to Type 5 ADC to output data collected from the ADC pins on the processor.
SOC: Start of Conversion
EOC: End of Conversion
SOC Trigger
Input Channels
Conversion channel ADCIN0
OK
Cancel
Help
Apply

ADC

In our MCU we have 4 ADC modules.

Which one?

Example: How to read the DC-Bus voltage (Power Supply)?

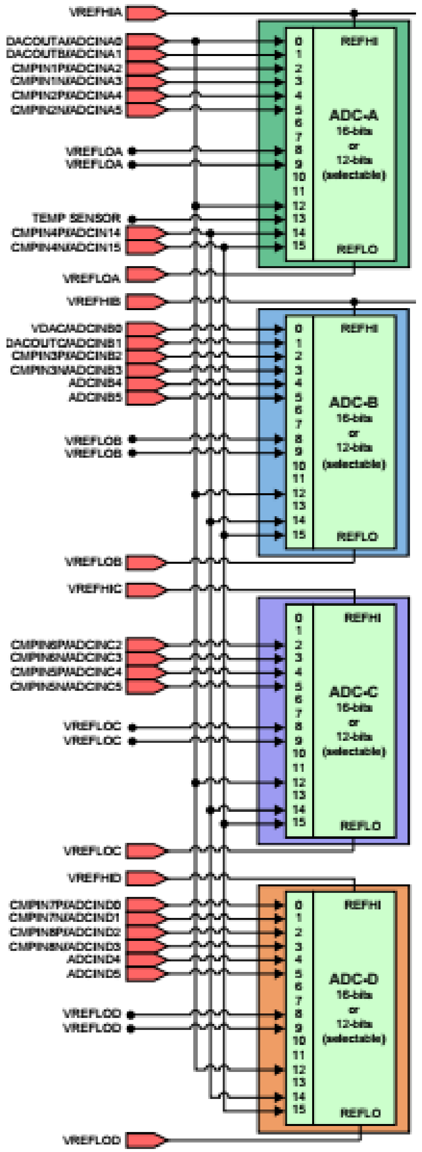


66	ADCINA5	CMPIN2N		VSEN_PVDD

ADCIN A 5

MODULE

INPUT CHANNEL



A

B

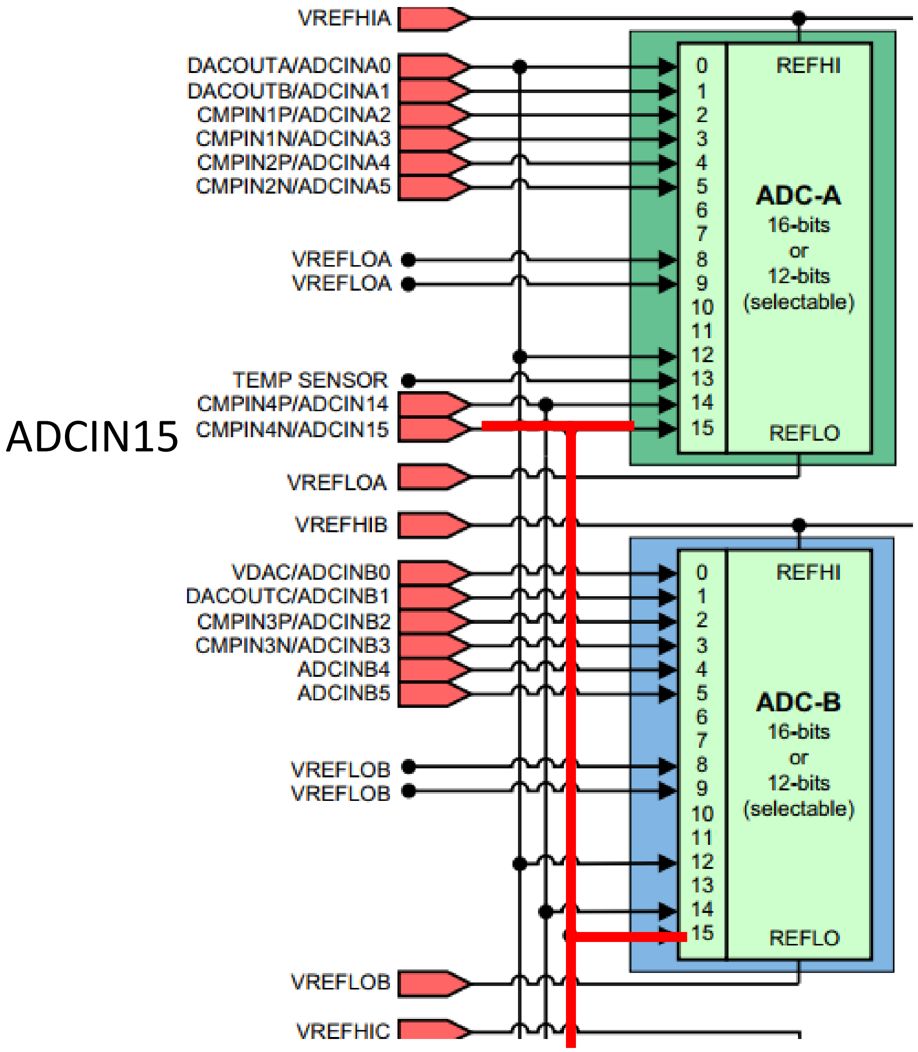
C

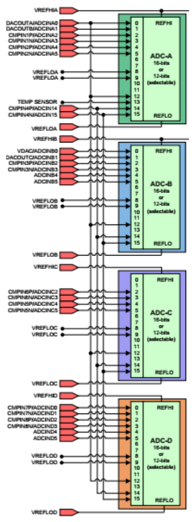
D

ADC

Some input channel are shared

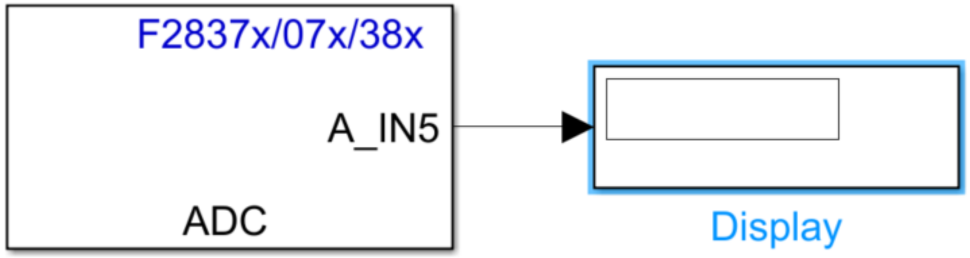
No module letter

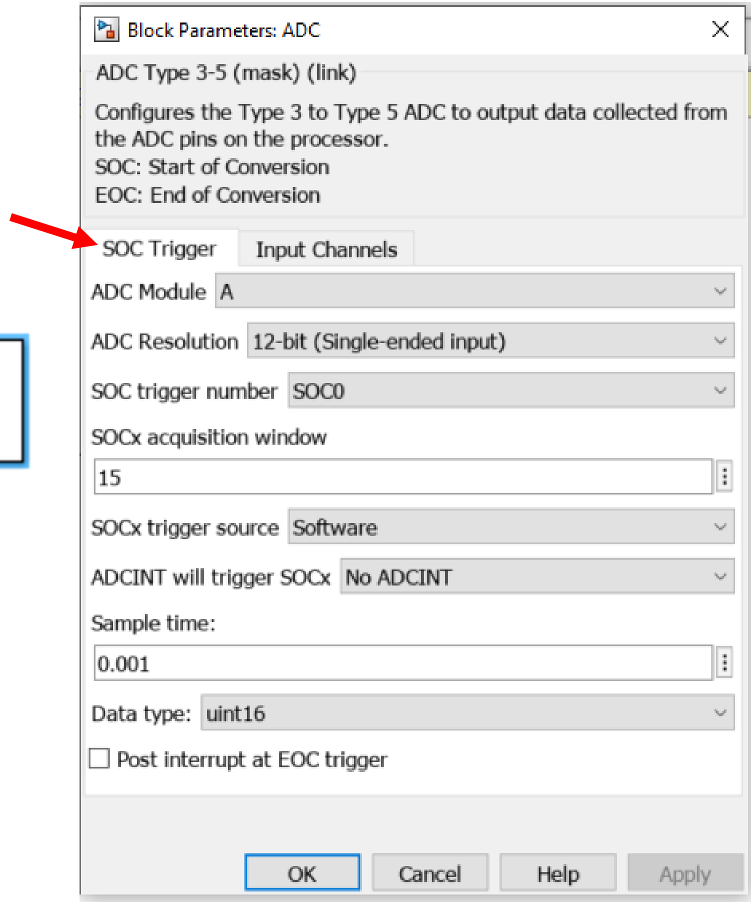




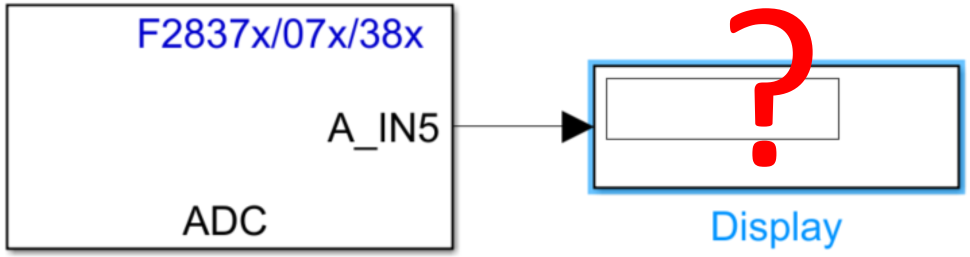
We want to know the supply voltage of our board

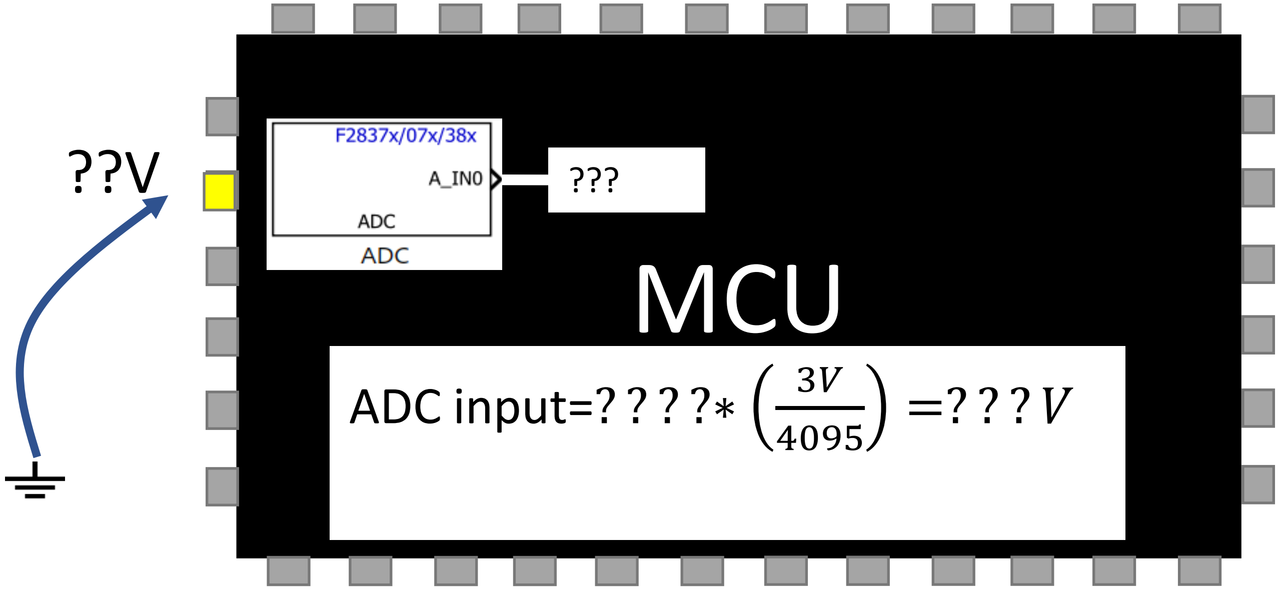
ADCIN A 5





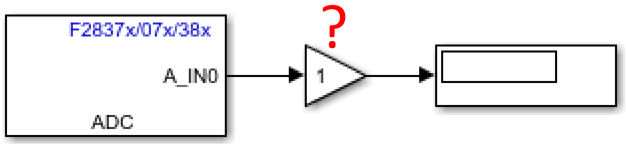
ADC





What is the input voltage?

You can put a gain to convert the value.

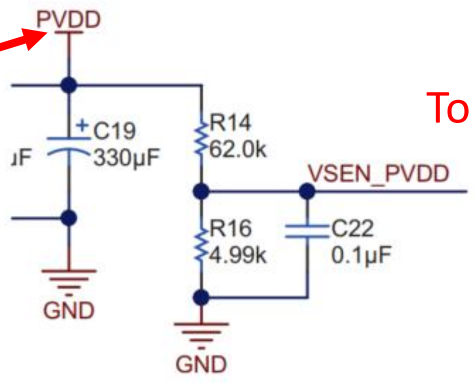


ADC VOLTAGE READING

5.1 *DC Bus and Phase Voltage Sense*

The BoosterPack has been designed with voltage sense circuits on the DC bus (PVDD) and each half-bridge outputs (phases A, B, and C). These circuits, shown Figure 9, consist of a voltage divider with a filtering capacitor to reduce high frequency noise on the ADC pins. These circuits have been scaled to support 4.4 to 45 V. The high-side resistors for the phase outputs are located near the motor output header (J4) while the low-side resistors and filtering capacitors are located near the ADC pins on the BoosterPack to LaunchPad header (J1) for improved noise reduction purposes.

Power Supply



To Microp.

Figure 9. Voltage Sense

$$VSEN\_PVDD = PVDD * \left(\frac{R16}{R14+R16}\right)$$

$$\boxed{VSEN\_PVDD = PVDD * 0.0745}$$

V=f(ADCValue)?

ADC

EXERCISE:

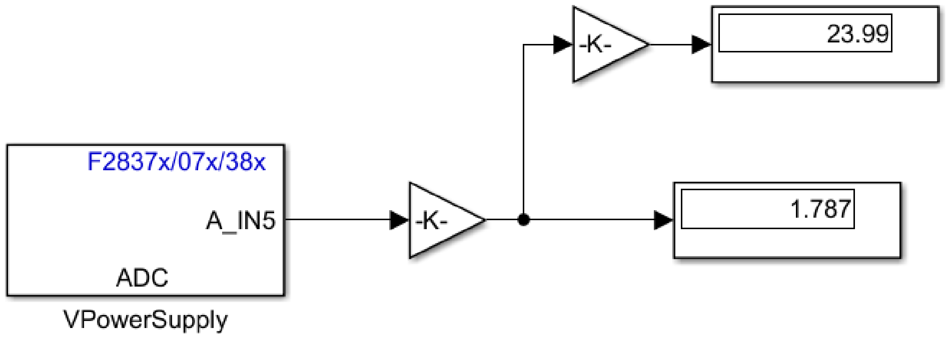
1: Read Vbus (VSEN PVDD)

Write a Simulink block (1 or more Gain) to convert the reading coming from the ADC into the corresponding voltage value.

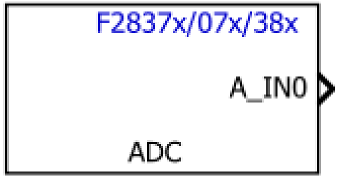
For the parameter, please edit an .m file with the values

```
%% Power Supply Voltage sense
ADCResolution = 2^12;
ADCCount=ADCResolution-1;
VrefADC=3;
GainVsense=4.99/(62+4.99);
……………………..
```

$$VSEN\_PVDD = \boxed{PVDD} * \left(\frac{R14}{R14+R16}\right)$$



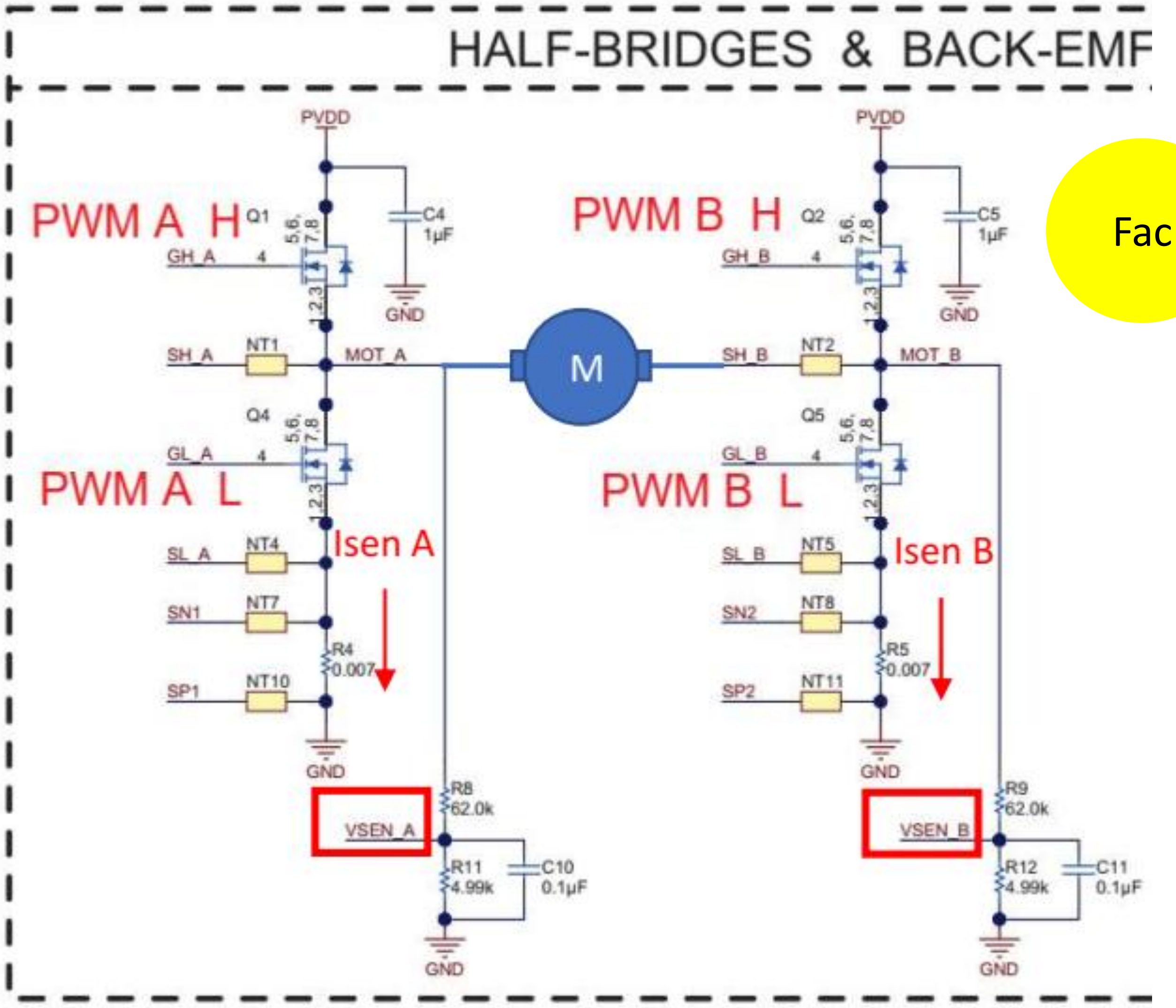
ADC



ADC

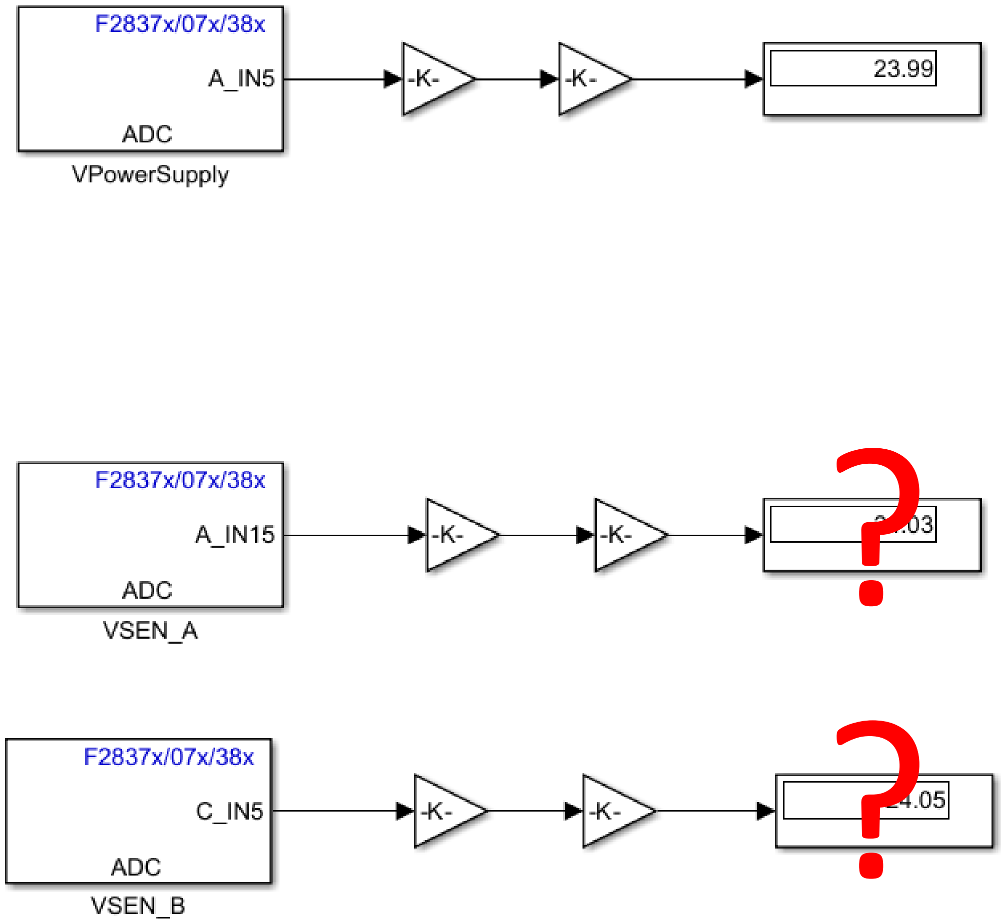
EXERCISE:

2: Read VSEN_A
Read VSEN_B



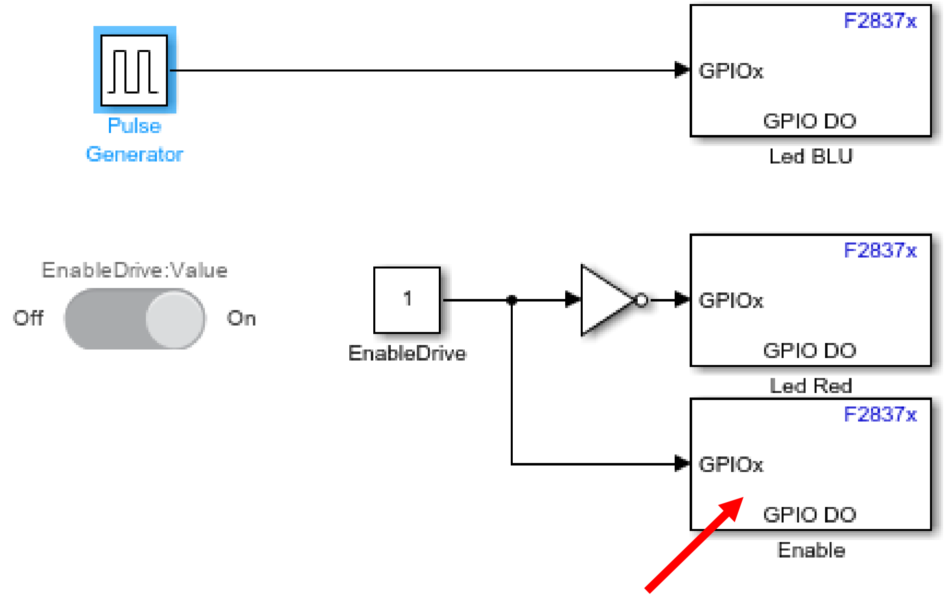
Fac

ADC



I need to drive my H-Bridge

ENABLE DRIVE



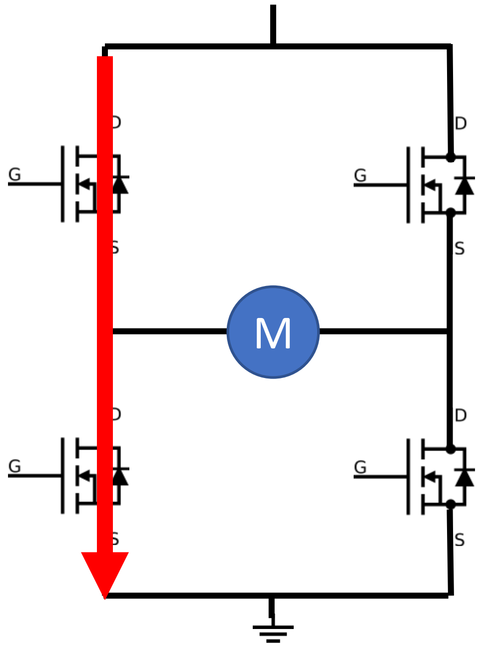
To Enable current reading you must enable the driver:

53	GPIO26	EN_DRIVER		SD2_D2 [1]

Use a Slider Switch to turn ON/OFF the Enable.
Use RED LED to indicate ENABLE DRIVER state:

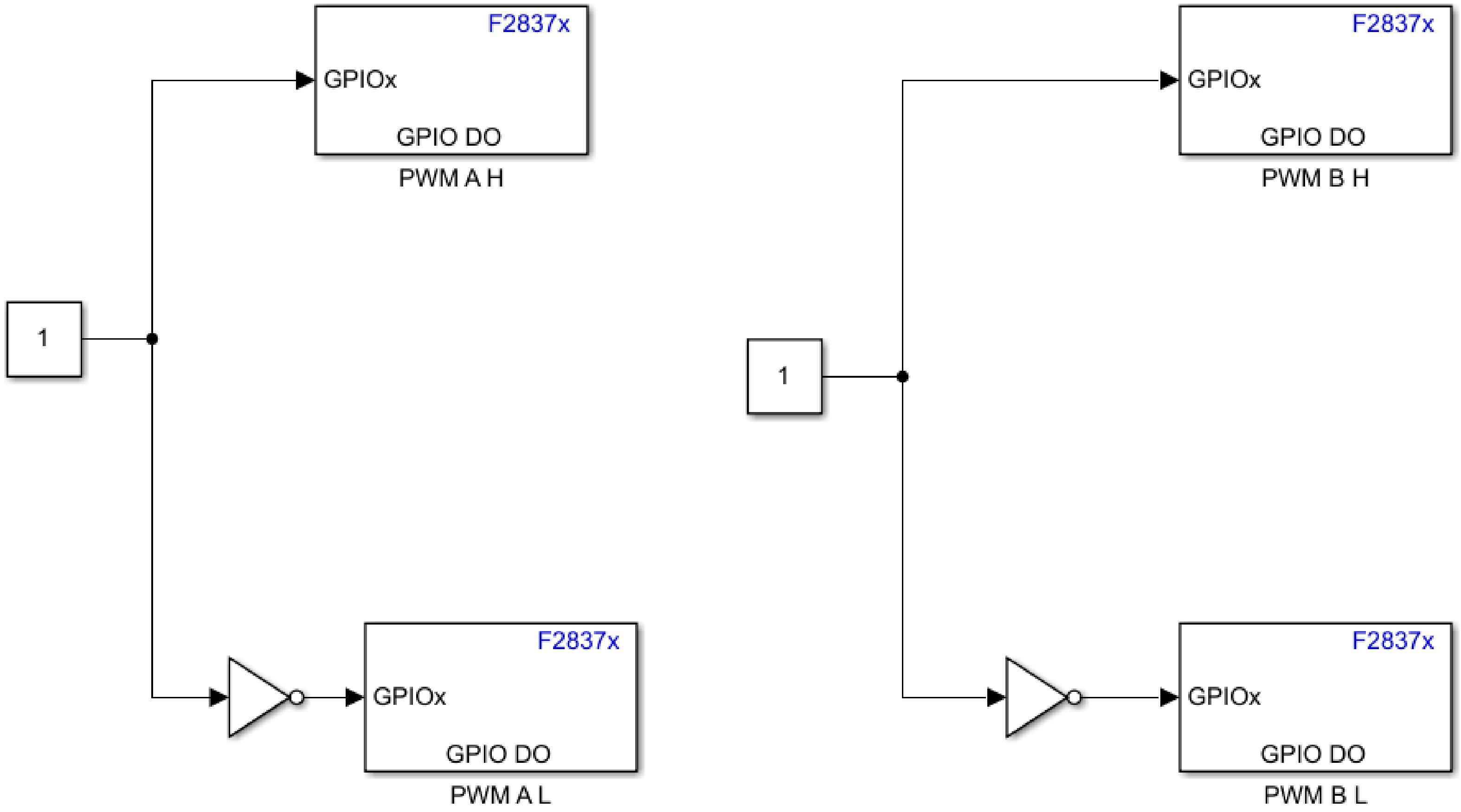
ENABLE = 0	LED RED = 0
ENABLE = 1	LED RED = ON

! ATTENTION !



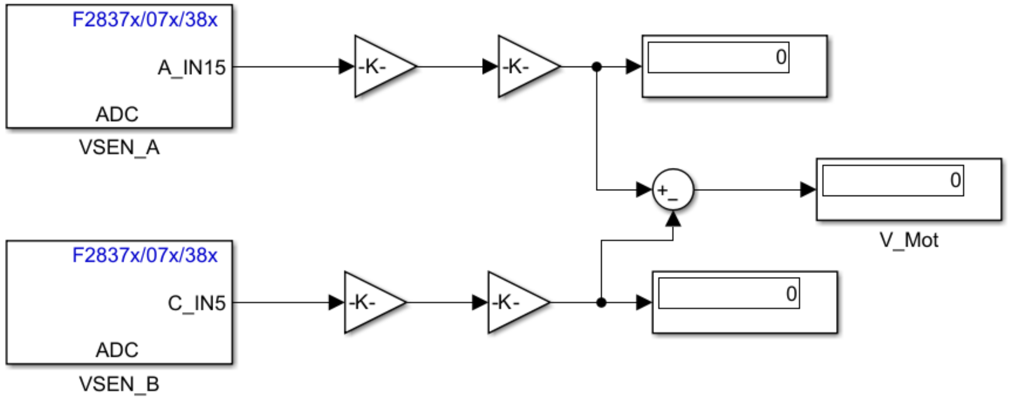
SHORT CIRCUIT

DRIVE H-BRIDGE



I can define **V_Mot = VSEN_A – VSE_B**

I obtain a positive or negative voltage, depending on the direction of rotation of the motor (clockwise anti-clockwise)



Now my motor is running, I cat measure current.

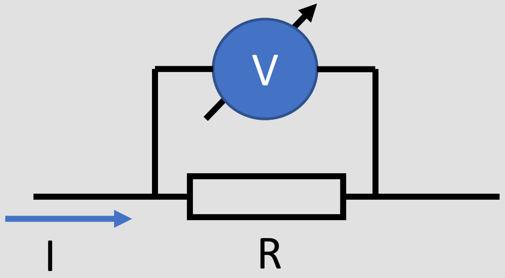
CURRENT SENSING

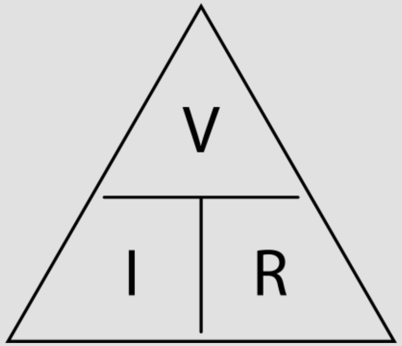
If we want to measure a current with our microcontroller, we must convert it into a voltage.

We can divide the measurement methods into two groups:

SHUNT RESISTOR

A current flow on a resistor and we measure the voltage dropout.

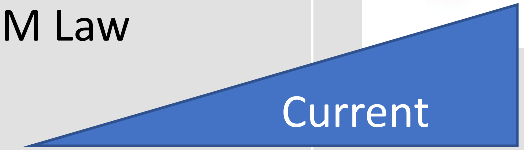




OHM Law

$$I = \frac{V}{R}$$

"ELECTROMAGNETIC FIELD" SENSOR



€€€€

CURRENT SENSING

The power dissipated on a resistance is:

$$P = I * I * R$$

If we have 10A and a resistance of 1Ω we have to dissipate 100W. <u>TOO MUCH POWER</u>

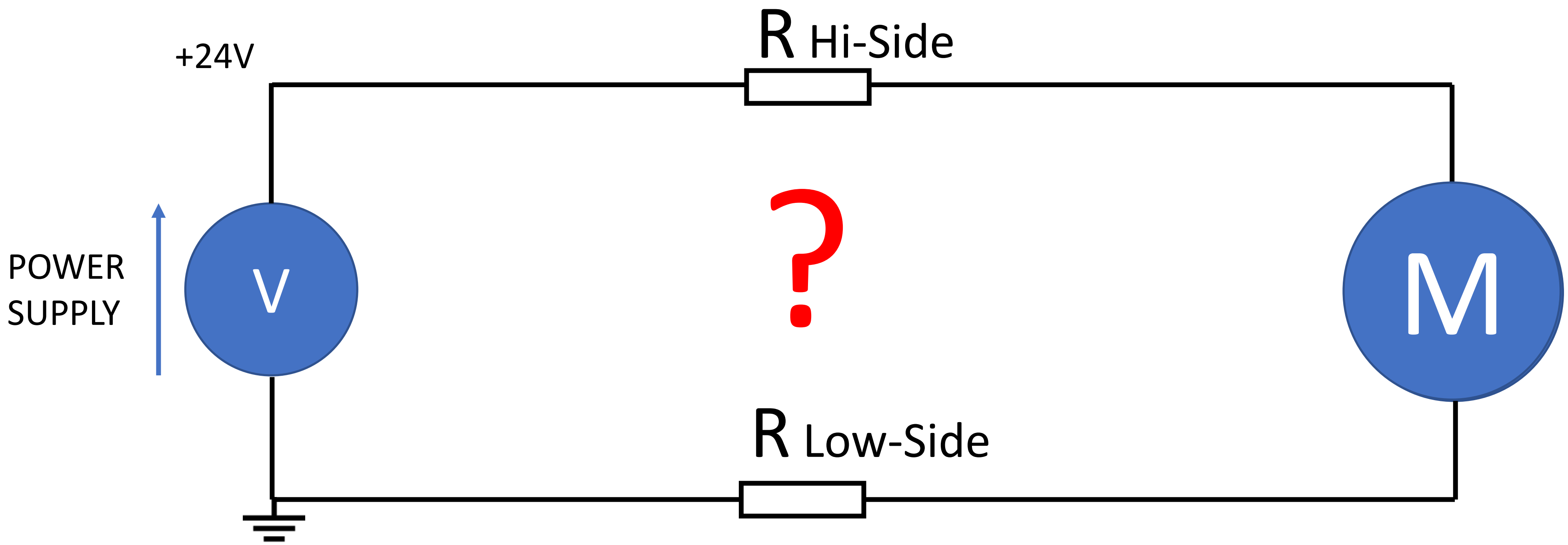
We use small resistance resistors. About 10mΩ

If we have 10A and a resistance of 10mΩ we have a voltage dropout of:

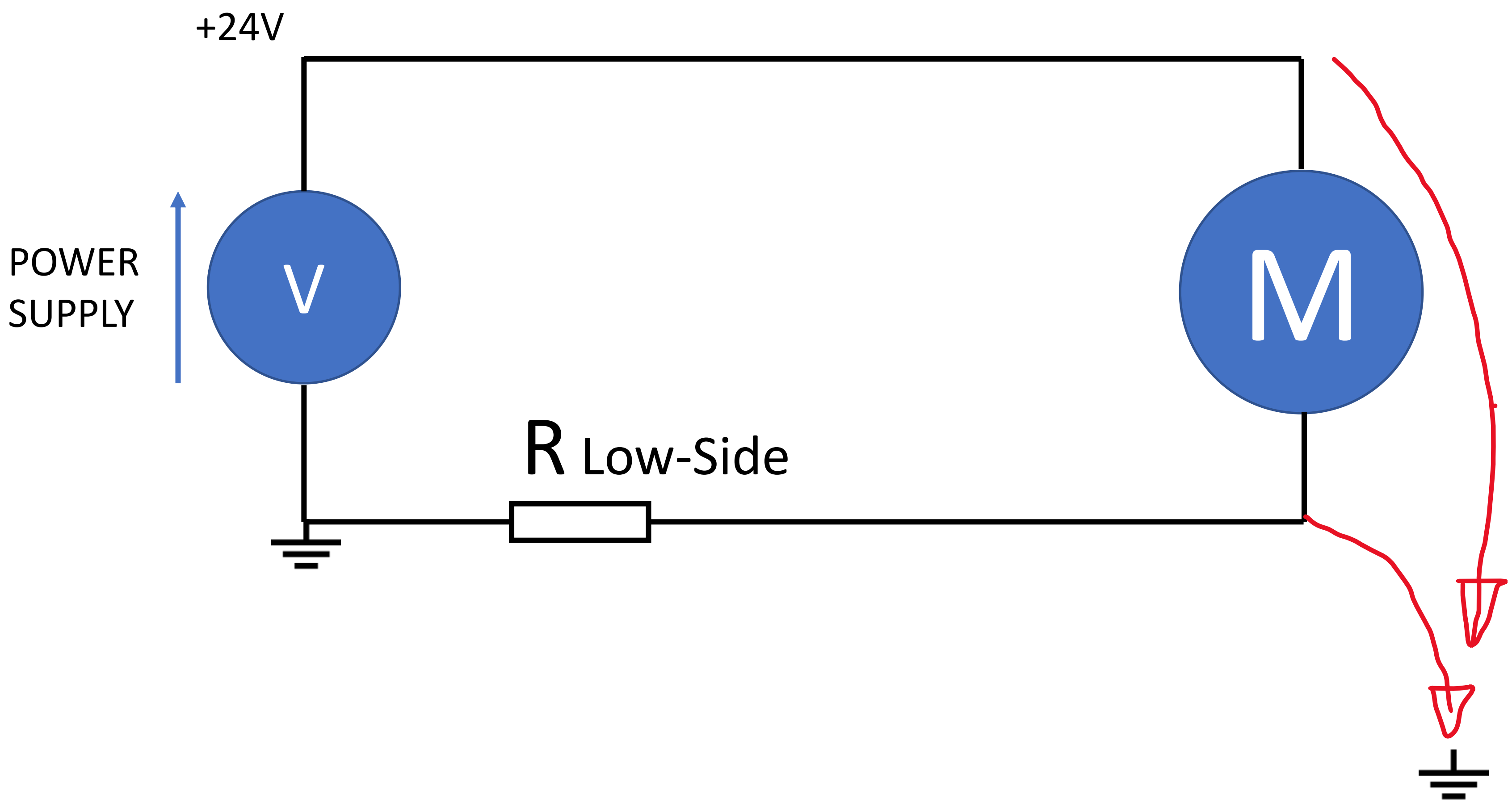
$$V = I * R = 10A * 0.010\Omega = 0.1\text{V}$$

A small value, we have to amplify it.

CURRENT SENSING



CURRENT SENSING – LOW SIDE



~~**FAULTS?**~~

IMPLEMENTATION

Single

Differential

GROUND DISTURBANCE

Yes

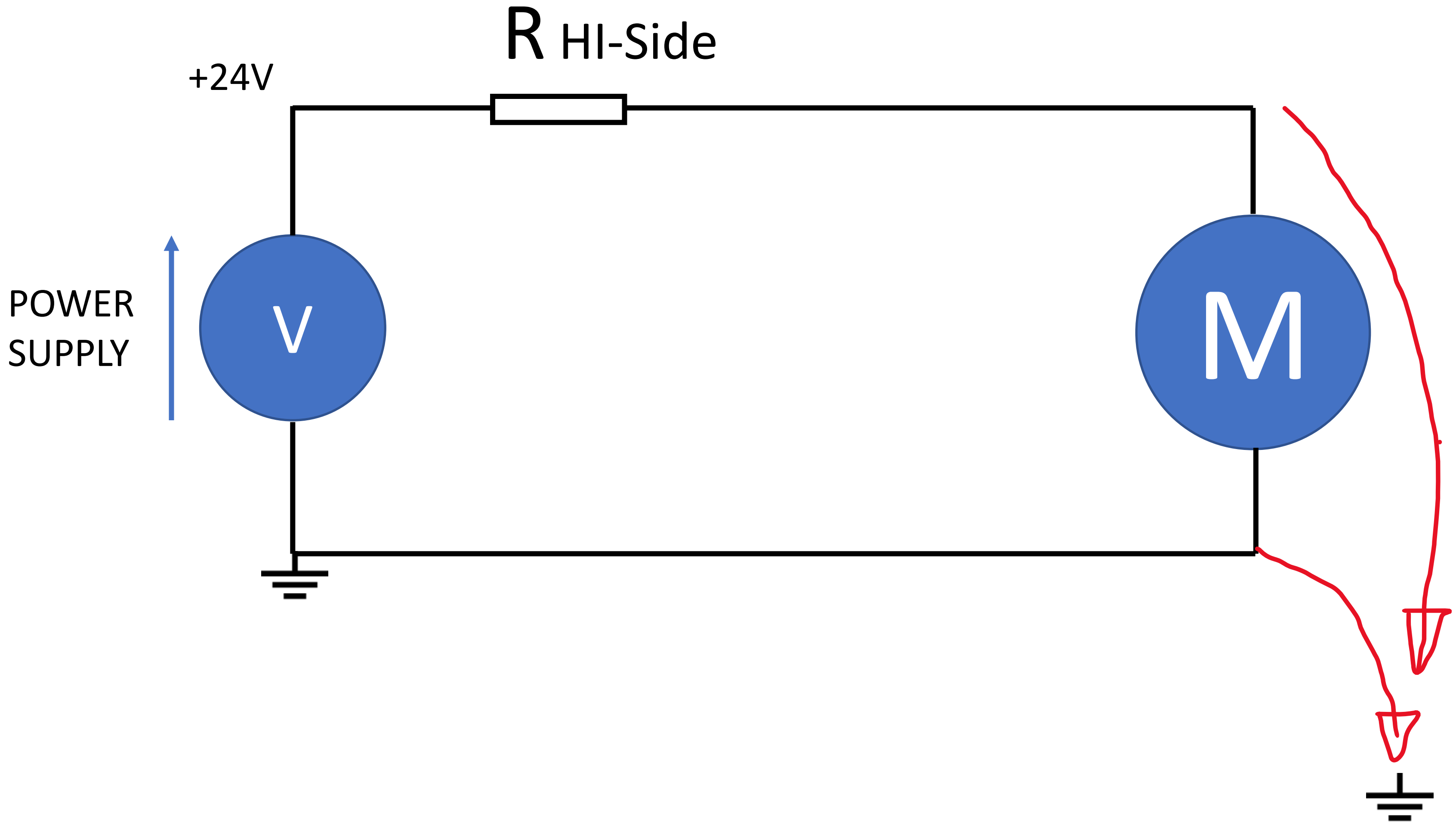
COMMON MODE VOLTAGE

Close to ground

CMRR

Low

CURRENT SENSING – HI-SIDE



FAULTS? ✓

IMPLEMENTATION

Differential

GROUND DISTURBANCE

No

COMMON MODE VOLTAGE

Close to Vdd

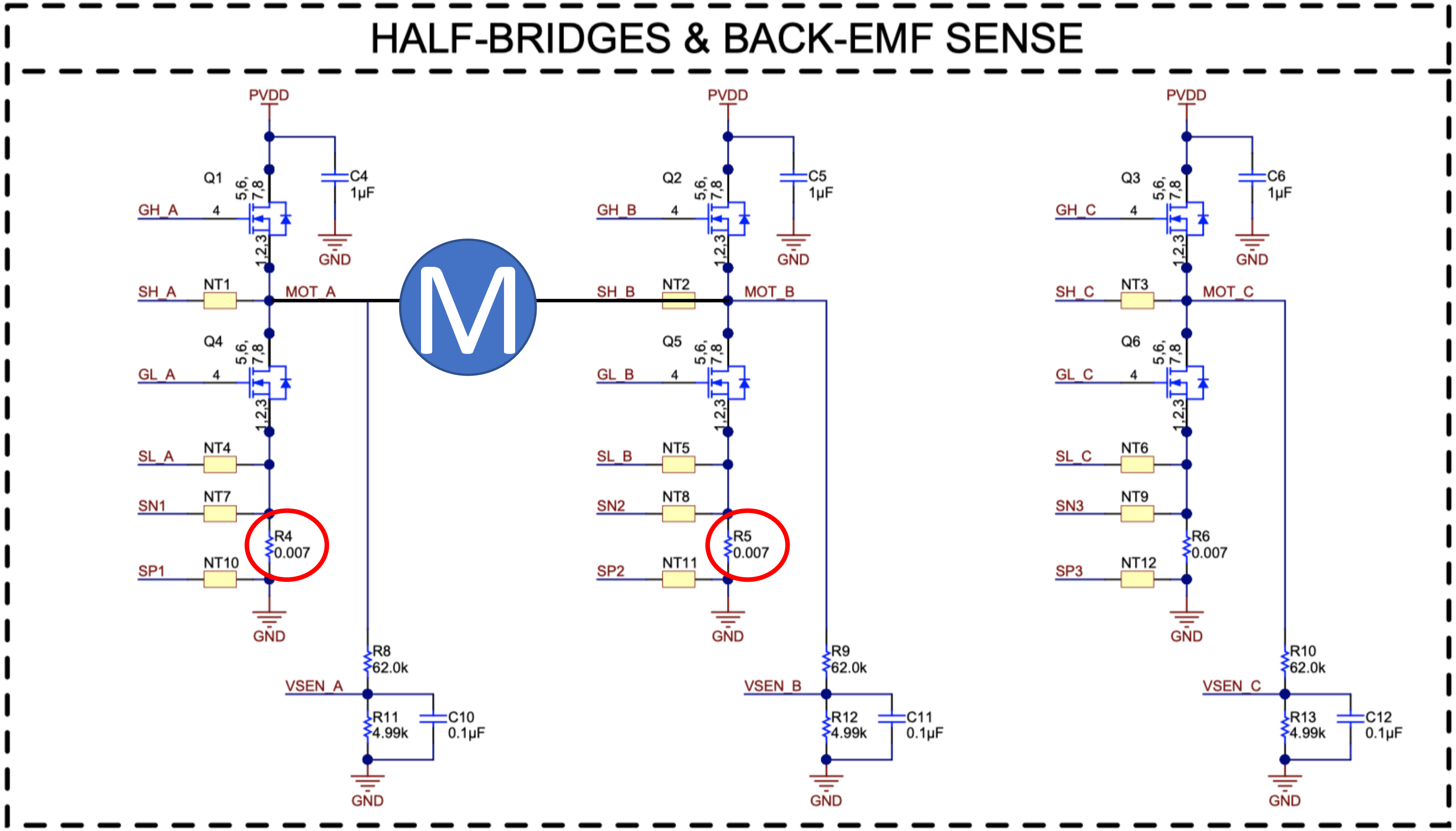
CMRR

HI

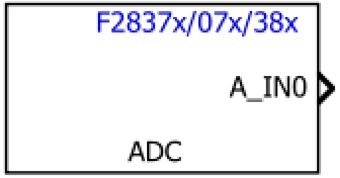
CURRENT SENSING – LOW-SIDE vs HI-SIDE

	LOW-Side	HI-Side
Fault (Lead shorts) Detect	NO	YES
Implementation	Single or Differential Input	Differential Input
Ground Disturbance	YES	NO
Common mode voltage	Closed to GND	Closed to Supply
CMMR (Common Mode Rejection Ratio)	Lower	Higher

CURRENT SENSING – **LOW-SIDE or HI-SIDE?**



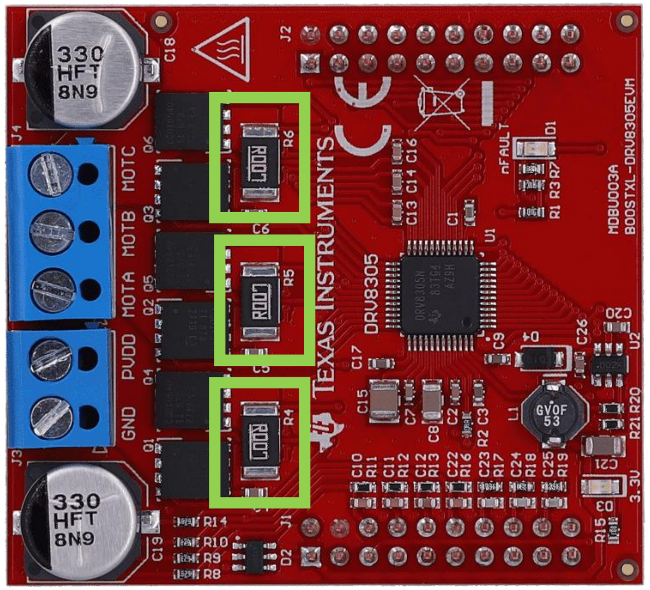
ADC



ADC

EXERCISE:

Measure ISEN_A

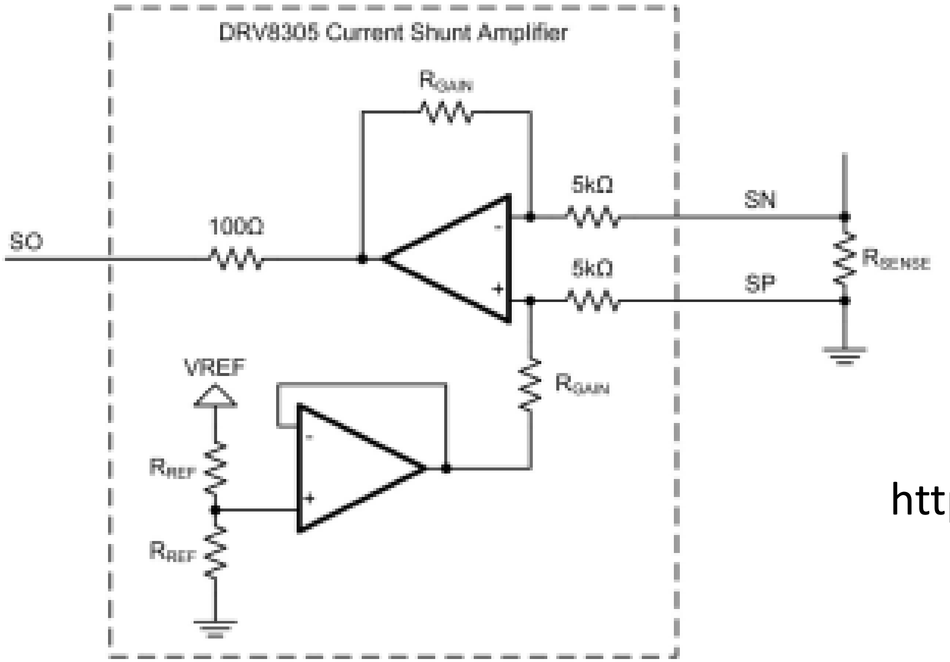


ADC CURRENT SENSING – OUR BOARD

Fac

5.2 *Low-Side Current Shunt Sense*

The BoosterPack has low-side current shunt sense for each half-bridge (phases A, B, and C). The current sense setup takes advantage of the DRV8305's triple shunt current amplifiers (phases A, B, and c). The configuration for the low-side sense is shown in Figure 9. The differential amplifier senses voltage across a 0.007-Ω power sense resistor with differential connections. The differential voltage is then amplified by 10 V/V and centered at 1.65 V to allow for sensing both positive and negative currents. The sense resistor has been scaled for 0- to 20-A peak currents.



https://www.ti.com/lit/pdf/slvuai8

ADC

CURRENT SENSING

Fac

Please write a Simulink model able to convert the reading coming from the ADC into the corresponding current value.

For the parameter, please edit an .m file with the values

```
%% Current Sense
R_shunt = 0.007;
……
```

Rsense=0.007Ω

Av = 10V/V

Voffset=1.65V

VREFADC=3V

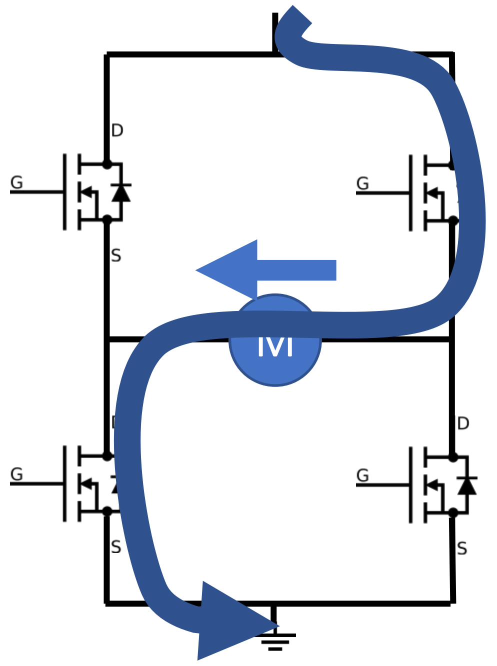
ADCResolution=12bit

I=f(ADCValue)?

ADC

Fac

If we want a current flowing in the circuit (ISEN A), we have to turn on the motor:



PWM B H = GPIO8

PWM A L = GPIO7

Fac

EXERCISE:

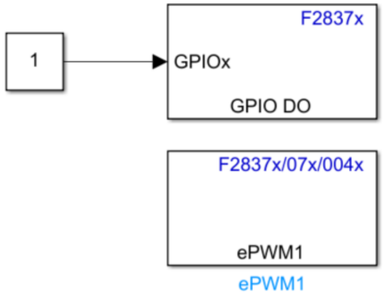
- Try with brake resistor
- Try to reverse motor rotation

PWM

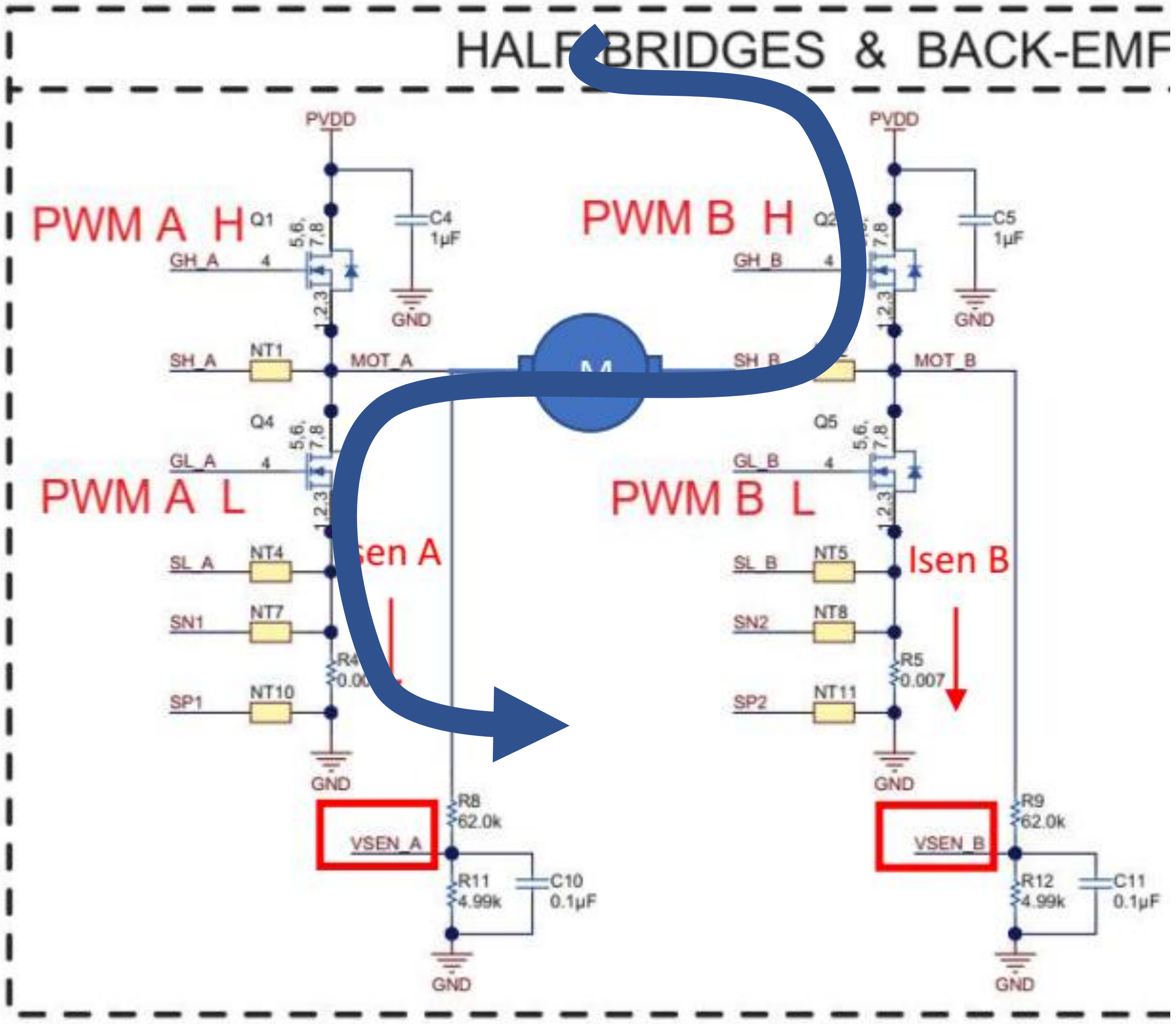
In order to have motor spinning, we can:

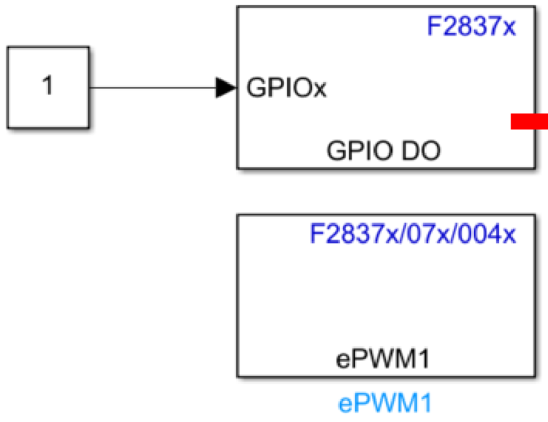
- Turn ON PWM BH
- PWM on PWM AL
- Enable Drive

PWM



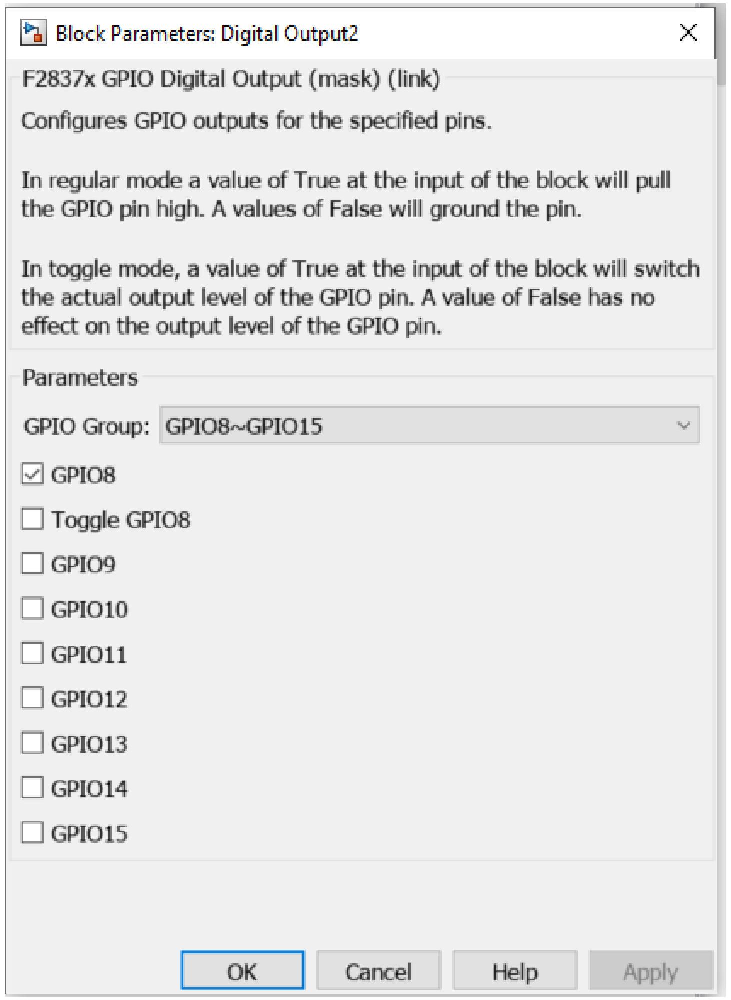
ENABLE

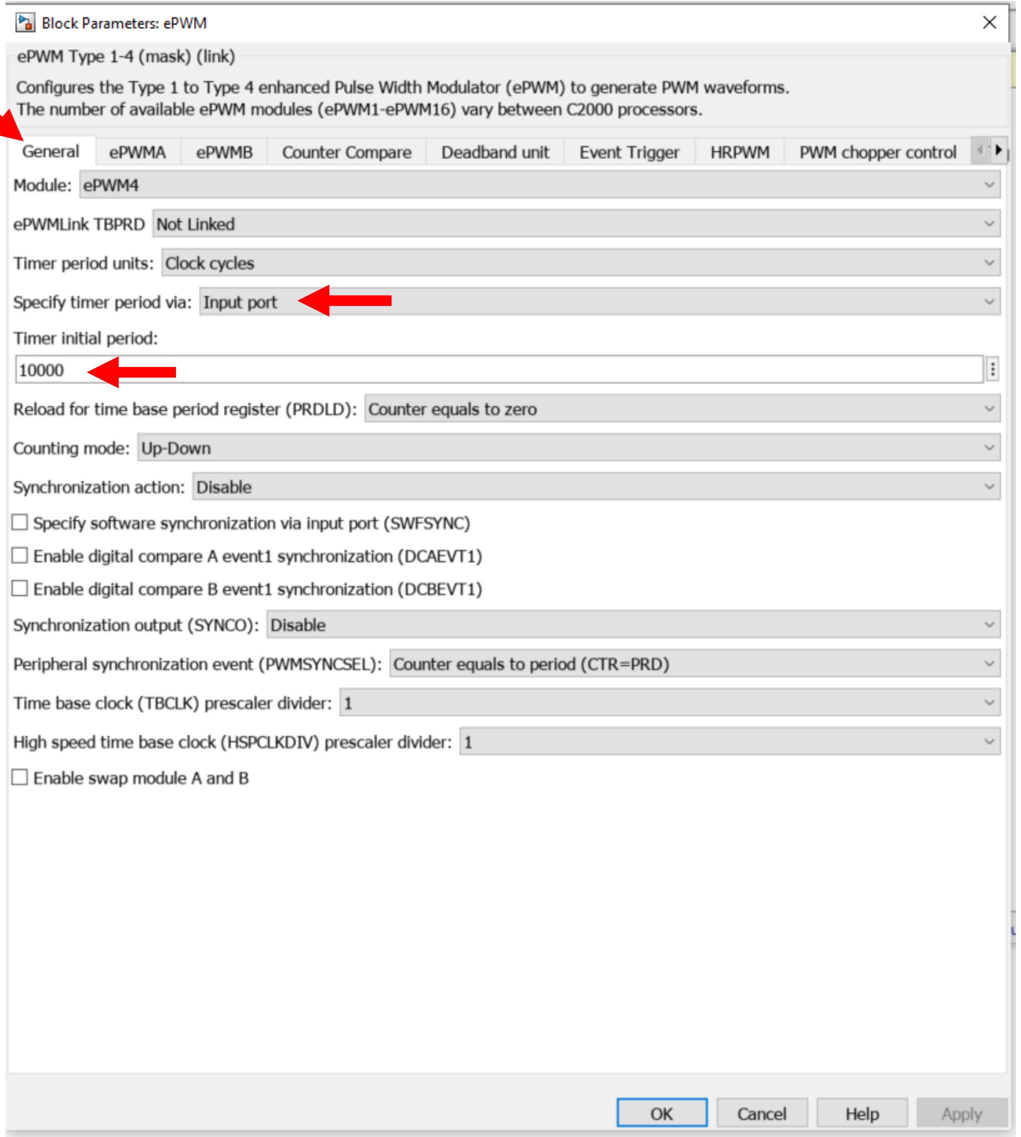


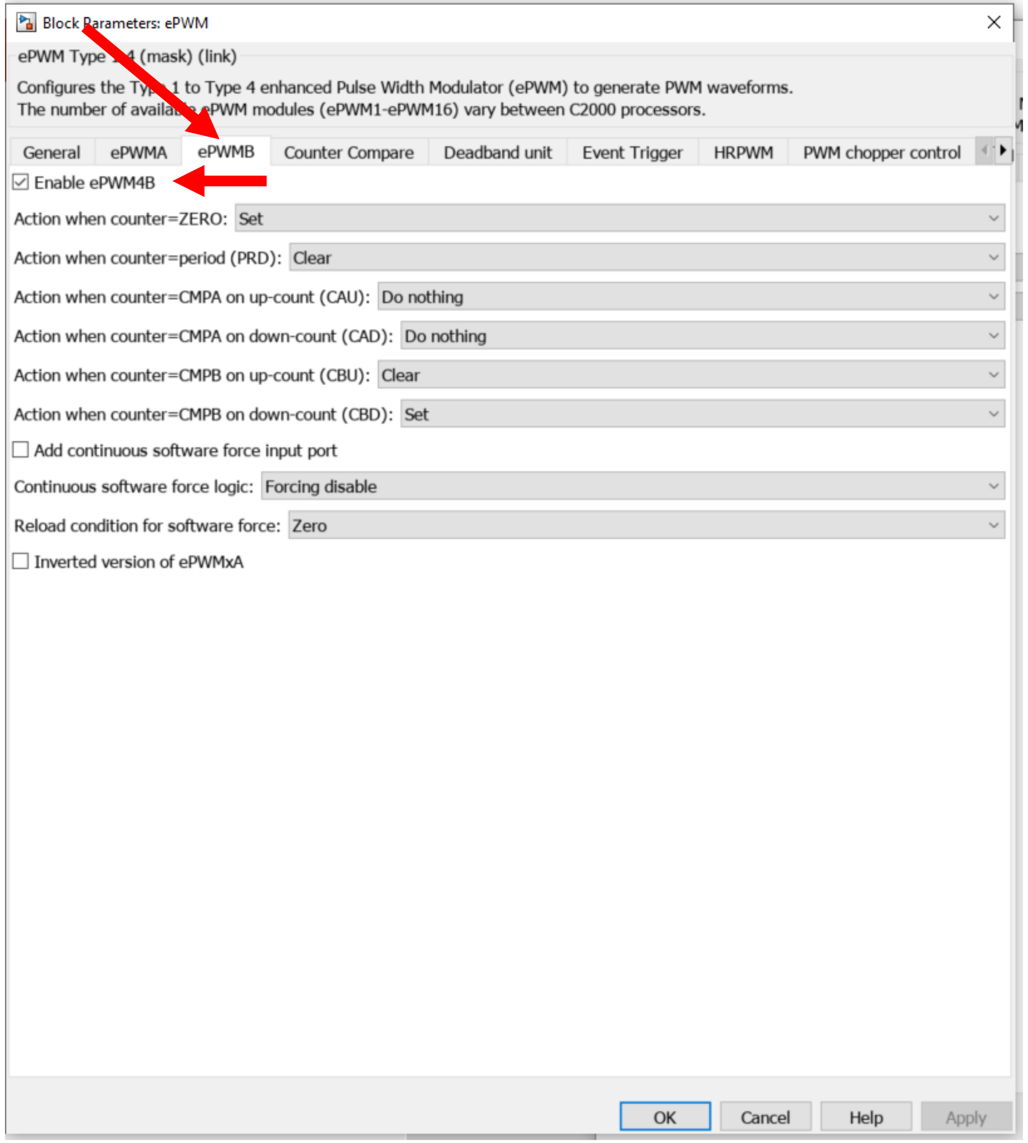


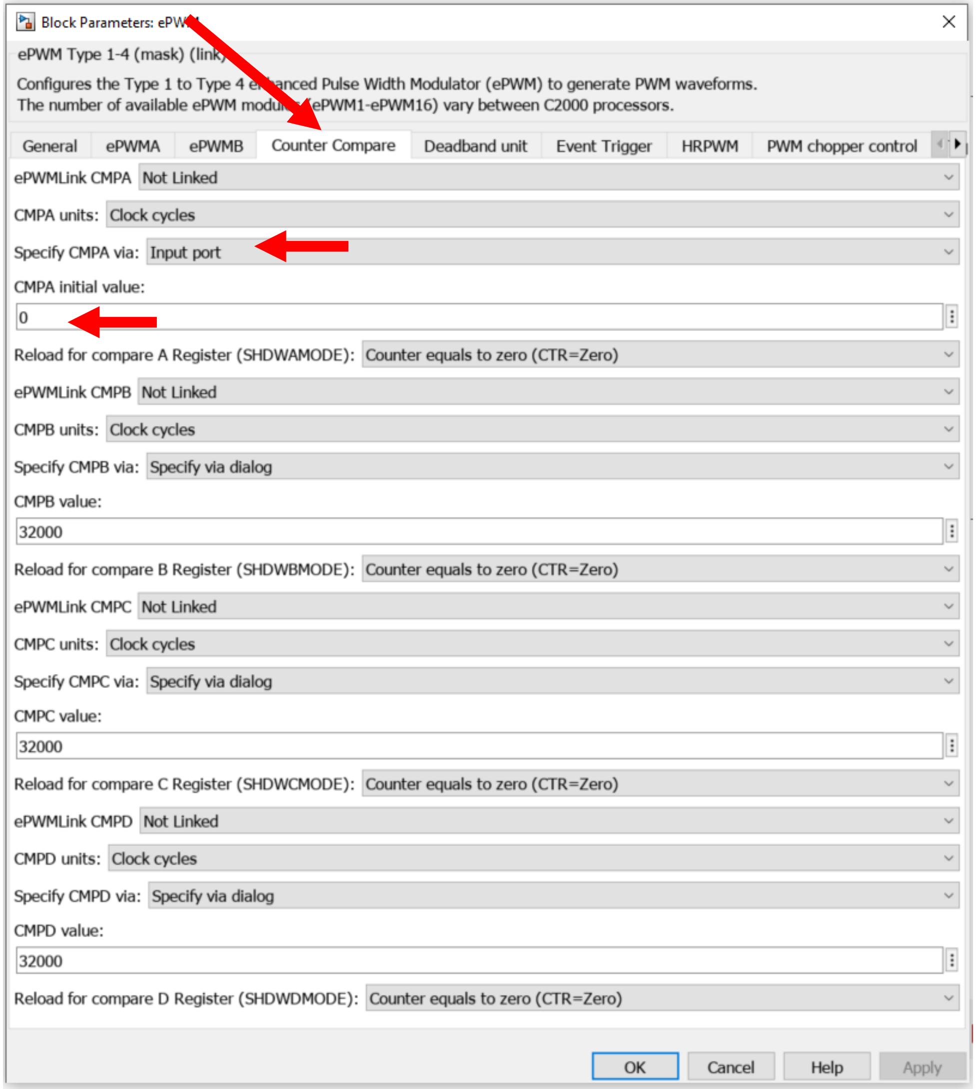
PWM BH -> GPIO???

PWM AL -> EPWM??

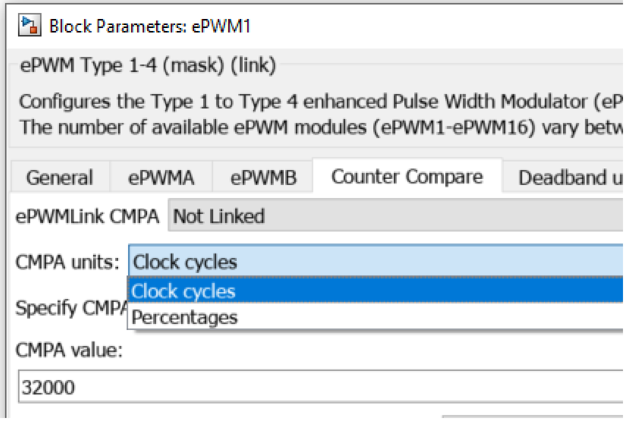


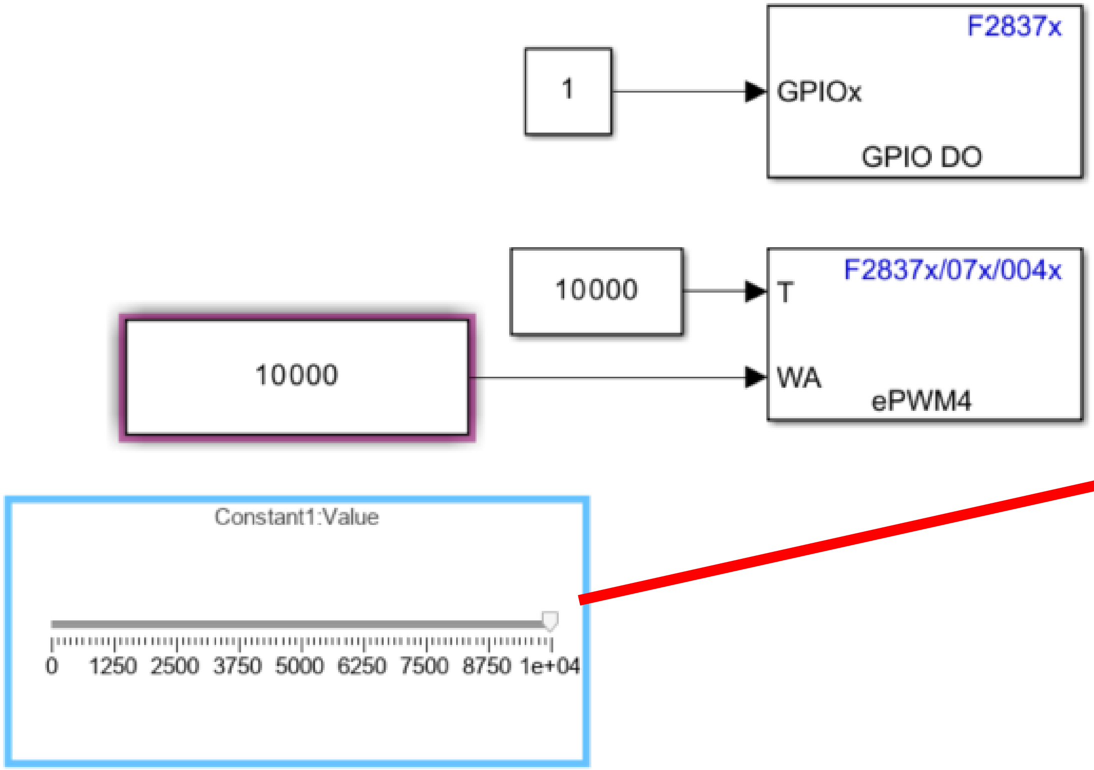

Block Parameters: ePWM
ePWM Type 1-4 (mask) (link)
Configures the Type 1 to Type 4 enhanced Pulse Width Modulator (ePWM) to generate PWM waveforms.
The number of available ePWM modules (ePWM1-ePWM16) vary between C2000 processors.
General
ePWMA
ePWMB
Counter Compare
Deadband unit
Event Trigger
HRPWM
PWM chopper control
Module: ePWM4
ePWMLink TBPRD Not Linked
Timer period units: Clock cycles
Specify timer period via: Input port
Timer initial period:
10000
Reload for time base period register (PRDLD): Counter equals to zero
Counting mode: Up-Down
Synchronization action: Disable
Specify software synchronization via input port (SWFSYNC)
Enable digital compare A event1 synchronization (DCAEVT1)
Enable digital compare B event1 synchronization (DCBEVT1)
Synchronization output (SYNCO): Disable
Peripheral synchronization event (PWMSYNCSEL): Counter equals to period (CTR=PRD)
Time base clock (TBCLK) prescaler divider: 1
High speed time base clock (HSPCLKDIV) prescaler divider: 1
Enable swap module A and B
OK
Cancel
Help
Apply

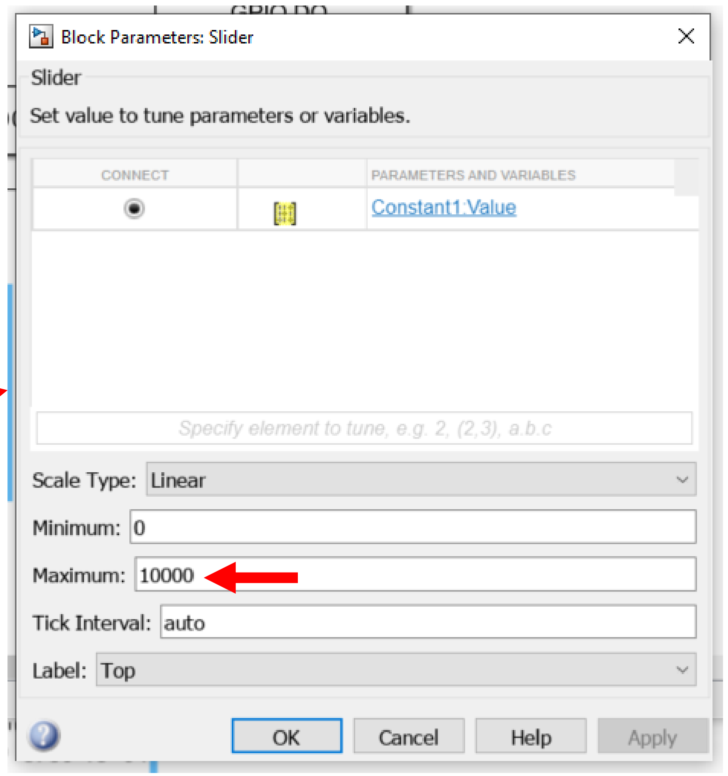

Block Parameters: ePWM
ePWM Type 1-4 (mask) (link)
Configures the Type 1 to Type 4 enhanced Pulse Width Modulator (ePWM) to generate PWM waveforms.
The number of available ePWM modules (ePWM1-ePWM16) vary between C2000 processors.
General
ePWMA
ePWMB
Counter Compare
Deadband unit
Event Trigger
HRPWM
PWM chopper control
Enable ePWM4B
Action when counter=ZERO: Set
Action when counter=period (PRD): Clear
Action when counter=CMPA on up-count (CAU): Do nothing
Action when counter=CMPA on down-count (CAD): Do nothing
Action when counter=CMPB on up-count (CBU): Clear
Action when counter=CMPB on down-count (CBD): Set
Add continuous software force input port
Continuous software force logic: Forcing disable
Reload condition for software force: Zero
Inverted version of ePWMxA
OK
Cancel
Help
Apply



I Can also set directly percentages




F2837x
1
GPIOx
GPIO DO
10000
F2837x/07x/004x
T
10000
WA
ePWM4
Constant1:Value
0 1250 2500 3750 5000 6250 7500 8750 1e+04


Block Parameters: Slider
Slider
Set value to tune parameters or variables.
CONNECT
PARAMETERS AND VARIABLES
Constant1:Value
Specify element to tune, e.g. 2, (2,3), a.b.c
Scale Type: Linear
Minimum: 0
Maximum: 10000
Tick Interval: auto
Label: Top
OK
Cancel
Help
Apply